

**Evaluating the Impact of AI-Driven
Large Language Models on
Teaching Practices in Further Education (UK)**

Author: Saadia Adnan

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Dedication:

This dissertation is dedicated to the cherished memory of my late, beloved mother, whose unwavering love, strength and guidance continue to inspire me every day. It is also dedicated to the honoured educators whose encouragement and support have been contributory in my journey of brilliance in the field of Data Science.

Acknowledgement:

I would like to begin by expressing my heartfelt gratitude to Allah for granting me the strength, patience, and perseverance to complete this research.

I am deeply grateful to my supervisor and the module leader for their invaluable guidance, advice, and encouragement whose expertise has been a keystone in shaping the direction and quality of this dissertation.

Thanks to my colleagues from the EdTech team at New City College. Their support has made a real difference along the way.

I would also like to thank the University of East London and Unicaf for awarding the scholarship, resources, and learning environment I needed to succeed in this program.

And most of all, thank you to my family and friends, whose patience, encouragement, and trust in me helped me stay grounded and keep going, even when things got tough.

Preface:

Writing this dissertation has been a journey filled with challenges, growth, and some rewarding moments. As a Lecturer in Further Education, I've learned how data can lead to informed decision-making in education.

Balancing my work with my studies wasn't easy but it has taught me how to juggle responsibilities and think more critically. I now understand the power of data to bring positive and meaningful changes in different sectors.

This dissertation is for anyone in education, industry, or other fields, who wants to explore how data can lead to smart decisions and solutions.

Looking back on this journey, I feel thankful for the chance to combine my love for education with my growing passion for data science.

Saadia Adnan

Chadwell Heath, London, United Kingdom

23rd March 2025

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List of Abbreviations

AI-Artificial Intelligence

LLM- Large Language Models

ML- Machine Learning

FE-Further Education

NCC-New City College

ETF- Education and Training Foundation

SLT – Senior Leadership Team

MIS – Management Information System

Ofsted – Office for Standards in Education

EDS – Educational Data Science

CM – Curriculum Managers

CPD – Continued Professional Development

POST – Parliamentary Office of Science and Technology

Abstract:

Artificial Intelligence (AI) tools, built on Large Language Models (LLMs) and Natural Language Processing (NLP) are changing the way further education (FE) works in the United Kingdom. These technologies offer new ways to personalise and adapt learning in modern classrooms.

This study explains how LLMs can be used within FE colleges, where instructors often work with diverse learners with limited time and resources. As an FE lecturer, I have used TeacherMatic in real-world lesson planning. I wanted to figure out how AI influences daily teaching practices.

A mixed-method approach is employed in this research, which combines academic usage data and survey feedback from colleagues who took part in an AI pilot initiative at New City College. It also analyses how people used the tool over time, what they thought about it before and after using it, and whether their engagement changed over time.

Finally, this dissertation offers realistic insights into how AI tools could be incorporated into FE teaching, not just in theory, but in ways that work in practice. It also accentuates the importance of the ethical application of AI and data privacy in FE. Overall, this investigation provides practical insights into the thoughtful and strategic implementation of LLMs in the United Kingdom's FE sector by examining the intersection of AI, data science, and education.

Chapter 1: Introduction

Data science has become a groundbreaking discipline in the modern era that transforms unprocessed data into actionable knowledge that we can act on. It is known as a valuable interdisciplinary science across industries and academic settings. In education, influence of data science is growing too, predominantly in areas like Learning Analytics, Machine Learning (ML), Artificial Intelligence, data science itself, and Natural Language Processing (NLP). Educational Data Science (EDS) includes a broad range of unstructured data analysis methods that help solve academic challenges, mainly using unique datasets (McFarland et al., 2021). However, the field is so broad and not clearly defined, making it difficult to determine what educational data science means in individual everyday practice.

Estrellado et al. (2020) describe education as a complex, multilayered field, which makes it challenging for a data scientist to work because there is no universal framework to guide this emerging discipline. Building on this, education can be metaphorically viewed as a multi-storied building with many rooms. This structure in the UK includes private and publicly funded schools covering several key stages, Further Education (FE) colleges offer vocational and higher-level courses, and Higher Education (HE) institutes provide degree programs. Some parts of the system like policy making, funding, or analytics are often hidden from the students, but these are prevailing ways that shape the education.

Fearns et al. (2023) state that educational professionals engage with data in different ways; finance staff plans to optimise tight budgets, university administrators develop metrics to determine career readiness, and education consultants analyse student performance and attitudes. These groups approach data with different priorities, so even though there is a lot of potential in using data science, there is not yet a defined way to apply it across the board. Practitioners are continuously working to find ways to integrate data science tools within

education systems. The reality that many teaching staff have limited or no experience collaborating with professionals who combine expertise in educational content, programming, and statistics analysis adds complexity (UK Parliament POST, 2024). Conway (2010) introduced a Venn diagram showing that data science intersects three areas: subject knowledge, programming skills, and statistical knowledge as below:

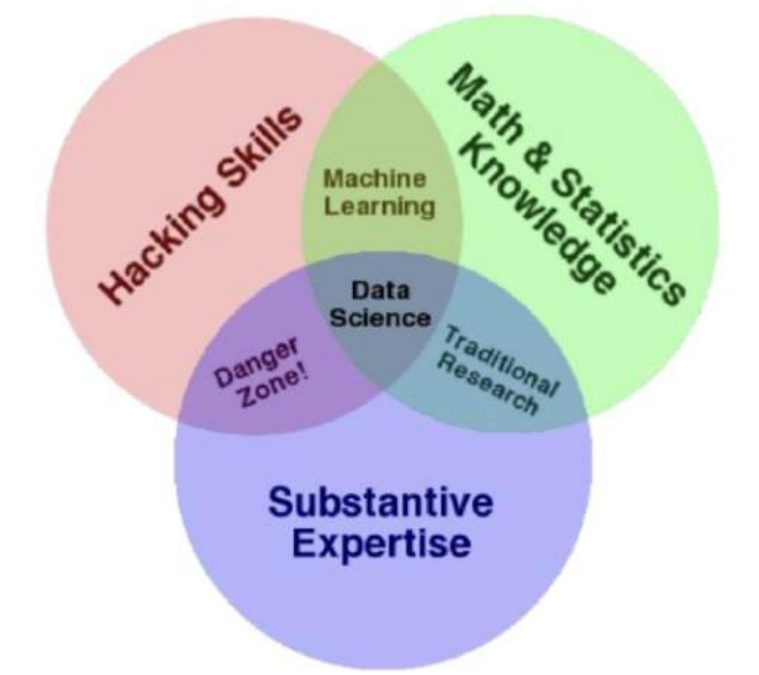


Figure 1. The Data Science Venn Diagram (Conway 2010)

Teachers might bring deep subject knowledge, but often lack formal training in coding or stats, so adopting these tools can be a real challenge.

Background of the Problem

Artificial Intelligence (AI) in education is reshaping teaching and learning models, offering new approaches to teaching and learning. However, the focus in academic research is on schools or universities. Further Education (FE) is often underrepresented. Dogucu et al. (2025) found that the current literature on AI and data science in education lacks cohesion. Therefore, there is a need for further research that examines real-world implementations of AI tools such as Large Language Models (LLMs) in FE settings. Gan et al. (2023) observe that LLMs are at the heart of broader discussions about the future of education. There is an increase in demand for personalised learning platforms that can meet individual student needs. LLMs can interpret and generate human-like language, which can be helpful in designing personalised and interactive educational experiences. OpenAI's ChatGPT and educational tools like TeacherMatic represents a significant step forward in how knowledge is accessed, shared, and organised (TeacherMatic, 2025). Despite their potential, research on the actual impact of these tools within the FE sector is still limited (Association of Colleges, 2024). That is the gap this study tries to explore: how are teachers using these tools, and do they make any meaningful difference to planning, delivery, or student learning?

Statement of the Problem Situation

Gen et al. (2023) and McFarland et al. (2021) both suggest that while Artificial Intelligence (AI) has shown promise in higher education and corporate training, its effectiveness in the Further Education (FE) is far less clear. That matters because the FE sector serves a wide range of learners with varied needs, learning styles, and goals, so it's important to understand whether Large Language Models (LLMs) truly enhance learning in this setting. Some students come straight from school, others return to education after years and juggle work or personal responsibilities. LLMs might help meet these needs but there is little research to confirm that. This research investigates these questions and bridges gaps by evaluating the capability of LLMs to enhance teaching and learning experiences in FE settings.

Purpose of the Study

This study evaluates recent implementation of Large Language Models (LLMs) within the UK's Further Education sector, focusing on their pedagogical value and impact on learning outcomes. This study:

1. Examines how LLMs help educators prepare teaching materials, save time, generate ideas, and matching materials to the curriculum.
2. Inspects AI adoption alongside the challenges and benefits of these tools.
3. Analyses educators' perspectives on AI's role in the learning environment.

Theoretical Framework

Saul (2024) explains Vygotsky's (1978) principle of the "more knowledgeable other" as a basic component of the *Zone of Proximal Development* (ZPD). ZPD is a theory that underlines the importance of guidance and learner growth beyond their independent capabilities. It is based on '*constructivist learning theory*', which believes that active, personalised engagement is necessary for the construction of knowledge.

The Artificial Intelligence (AI) based TeacherMatic tool is seen as an auxiliary tutor designed to assist educators and students within this framework. By automating and optimising resource creation, lesson plans, and content delivery, TeacherMatic supports Vygotsky's (1978) ZPD by offering targeted support that improves how teachers teach, and students learn.

To evaluate the depth and effectiveness of TeacherMatic's integration into teaching practices, this study applies the SAMR (Substitution, Augmentation, Modification, and Redefinition) Model. This model is used to decide whether the tool is just replacing traditional methods or is tactically transforming and enhancing teaching practices (Hamilton et al., 2016).

Importance of the Study

The Artificial Intelligence-based TeacherMatic tool has the potential to bring big changes in the Further Education (FE) sector by helping with sector-specific challenges. It can possibly help manage learning, keep students engaged, and reduce the amount of teachers' workload.

However, critical ethical issues related to AI adoption require further research, ensuring the technology does not have unfair biases, protecting student data, and maintaining a good balance between technology usage and the side of education. The findings aim to guide practitioners and policymakers in responsibly and effectively leveraging AI tools within FE.

Scope and Limitations of the Study

The study focuses on one Artificial Intelligence-based tool and how it is used within one Further Education college in the United Kingdom. It tests how TeacherMatic supports teaching tasks such as planning lessons, making resources, and delivering content but does not cover the technical details of how the AI works.

The limitations include probable biases in TeacherMatic's outputs, ethical concerns surrounding data use and privacy, and the challenge of balancing AI help and human-led teaching. While the study aims to offer practical insights, its findings can possibly only provide limited solutions due to the focus on one FE setting.

Overview and Rationale of the Study:

The use of data science and AI-powered tools like TeacherMatic is reshaping how teaching and learning happen in the Further Education (FE) sector. TeacherMatic helps educators plan lessons, create resources, and respond more effectively to individual learner needs.

This study investigates how TeacherMatic might transform teaching and learning by enhancing student engagement, improving learning outcomes, and reducing administrative workloads. Focusing on its implementation in FE, the research aims to bridge the gap between AI's theoretical promises and its practicality in everyday teaching.

Research Topic

Evaluate the Impact of AI-driven Large Language Models on Teaching Practices in Further Education in the UK

Research Question

Could TeacherMatic be employed effectively in Further Education to enhance the teaching and learning experience?

Research Hypothesis

Alternative Hypothesis (H_1): Using TeacherMatic in FE leads to better teaching, higher engagement, and improved learning outcomes.

Null Hypothesis (H_0): Using TeacherMatic in FE doesn't make a significant difference to teaching quality, student engagement or learning outcomes.

Chapter 2: Literature Review

There are existing theories on the use and implementation of Artificial Intelligence (AI) in education along with the pedagogical frameworks to guide its integration in Further Education (FE).

Data Science in Education

In education, data science is about using methods to study and understand information about teaching and learning. According to Estrellado et al. (2020), there are two main roles of data science in education., to apply data science techniques to educational data, and to include data into the curriculum. Data helps track long student absences, pick students who may need extra support, and to check if teaching methods are working well. McFarland et al. (2021) noted that these insights can help improve education overall. Predictive analytics can also help forecast how students will perform and help institutions use their resources better. Pesonen et al. (2018) say predictive models are useful in this way, and Fearn's et al. (2023) explain why data skills are important for the current job market. In FE, teachers and managers use advanced data methods such as inferential statistics and predictive modelling to understand students better, including their different learning levels and backgrounds through different applications available to them. Dougucu et al. (2025) proposed that these methods help teachers face challenges in education. Data Science is also the origin of AI tools like TeacherMatic, which uses data to create personalised learning and support teachers. The Association of Colleges (2024) and TeacherMatic (2025) both describe that the AI platform can immensely support teachers. Holmes et al. (2019) also explained that AI can improve education by building on data science.

Artificial Intelligence (AI) in Education

Gajjar (2023) explains in the UK Parliamentary Office of Science and Technology (POST) briefing that Artificial Intelligence technologies are underpinned by ‘machine learning’, which works by finding patterns in training data and using them to inform the processing of new data to generate outputs. Large Language Models (LLMs) are Foundation Models that carry out text processing. AI is like a smart computer system designed to perceive its surroundings, think, and perform tasks that affect the environment. Holmes et al. (2019) describe AI as a branch of computer science that uses algorithms, Machine learning (ML), and Natural Language Processing (NLP) to replicate human cognition, such as decision-making and problem-solving. In education, AI helps automate simple tasks, provides students personalised learning, and offers teachers useful information from data. Machine learning, deep learning, and generative AI are all connected in layers (IBM, 2024). Here is a timeline of their development:

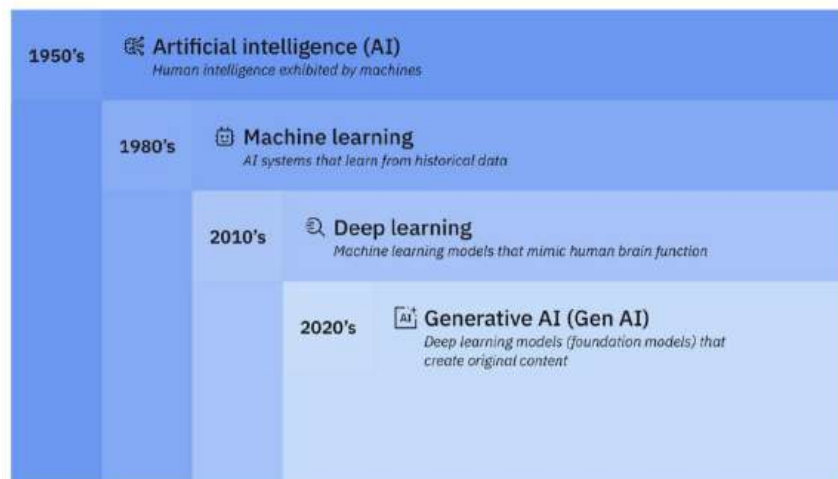


Figure 2: How AI, ML, deep learning, and Gen AI are related. (IBM, 2024)

There are different types of AI depending on their capacity:

Narrow AI is designed to handle specific tasks. It includes marking assignments, creating lesson plans, or supporting teachers with routine work. Tools like TeacherMatic are good examples of this type as these are designed for education-focused tasks (TeacherMatic, 2025).

General AI refers to a broader kind of system that can perform human-like intellectual tasks. However, this type is still theoretical and has not been achieved yet (Holmes et al., 2019).

Super AI is a future concept where machines could potentially surpass human intelligence. Like Gen AI, this idea exists mostly in speculation and theory at this point (Holmes et al., 2019).

As intelligent tools are becoming more common in schools and colleges, there is a greater need for technology that is simple to use. Tools that are simple to use, can adapt to larger groups of students, and offer tailored learning experiences (McFarland et al., 2021).

Natural Language Processing (NLP) and Large Language Models (LLMs)

One powerful capability of these tools is understanding and generation human-like language. This is made possible through Natural Language Processing (NLP). NLP allows systems to process written or spoken language in a way that mimics human understanding. Building on NLP, Large Language Models (LLMs) use deep learning to produce responses that are relevant and clear. These models are trained on large amounts of text and can create responses that suit our needs. For example, Gan et al., (2023) discuss that LLMs generate coherent and context-aware content. Kamath et al. (2024) highlight their use in improving the quality and relevance of educational materials.

LLMs in Education

Humans usually don't rely on just one way of understanding the world, they use their senses like sight, sound, and touch to gather information and respond to the stimulus accordingly. Similarly, innovative tools in education are becoming more and more precise in handling different tasks. Some of the language tools can work with voice also. These tools are known as Large Language Models (LLMs) in simple language which help computers understand and use natural human language. LLMs use deep learning methods to write content. For example, GPT-4 can navigate users and give answers based on their prompts and questions (Kamath et al., 2024). This form of Artificial Intelligence is helping improve education by providing personalised and targeted learning. The tools can give precise feedback and support one-to-one sessions. This is also helpful in creating materials according to the learners' needs. It allows learners to get extra help when teachers aren't around (Baker & Smith, 2019).

The UK Department of Education (2024) has published a recent report which confirms that many teachers are starting to use generative tools to personalised learning and save time on repetitive tasks. This means more time and energy for teaching.

LLMs such as TeacherMatic help with lesson plans and routine tasks allowing teachers more time to focus on students in live classrooms. Research has shown that these tools can make a difference but how and to what extent remains unexplorable. For example, one study showed that students were 30% more engaged in tutoring sessions led by LLMs. Students found LLMs support better for tricky STEM topics (Chen et al., 2023). This means that LLM could be a great support in Further Education (FE), where students learn at different speeds and come from varied backgrounds (El Fathi et al., 2025).

Applications of LLMs in Further Education (FE)

Further Education (FE) institutions serve a diverse group of learners, from teenagers continuing their studies to adults returning to education after a break. This diversity presents real challenges for teachers, who need to adapt lessons to suit different academic backgrounds, learning paces, and personal goals. On top of that, many FE colleges face tight budgets and staff shortages, making it difficult to provide personalised support to every student. To tackle these challenges, AI-powered tools are being adopted to streamline planning, marking, and resource creation. As Bolla (2025) states these AI tools are seen to improve education quality despite limited resources.

While these applications highlight the transformative potential of state-of-the-art LLM models, challenges remain. Despite these benefits, concerns remain about fairness and accessibility, as LLMs may overlook cultural or linguistic differences without careful oversight, potentially affecting equitable learning experiences (Yan et al., 2023).

TeacherMatic: An Ultimate Tool for Educators

TeacherMatic is a digital platform designed to support teachers and other staff in colleges. It helps with a range of tasks e.g. planning, teaching and assessment activities, etc. Since it was launched in April 2023, it has been adopted by over 60% of Further Education colleges. On average, teachers save 3.8 hours of work per week (Association of Colleges, 2024).



Figure 3: The revolutionary AI Platform created by educators for educators.

TeacherMatic (2025) is more relevant to Further Education teaching due to its development process. This transformative tool is designed with direct input from over 300 educators. As a result, it includes features that have an ability to automate and customise lesson plans, quizzes, and schemes of work. It also enables teachers to focus on developing connections with students and enhancing their learning experiences during their time in the classroom.

TeacherMatics's goal is to improve teachers' work-life balance while maintaining educational standards irrespective of the setting where it is being used. By saving time and energy, it

empowers educators to dedicate more attention to their students and enjoy more personal time with family and friends.

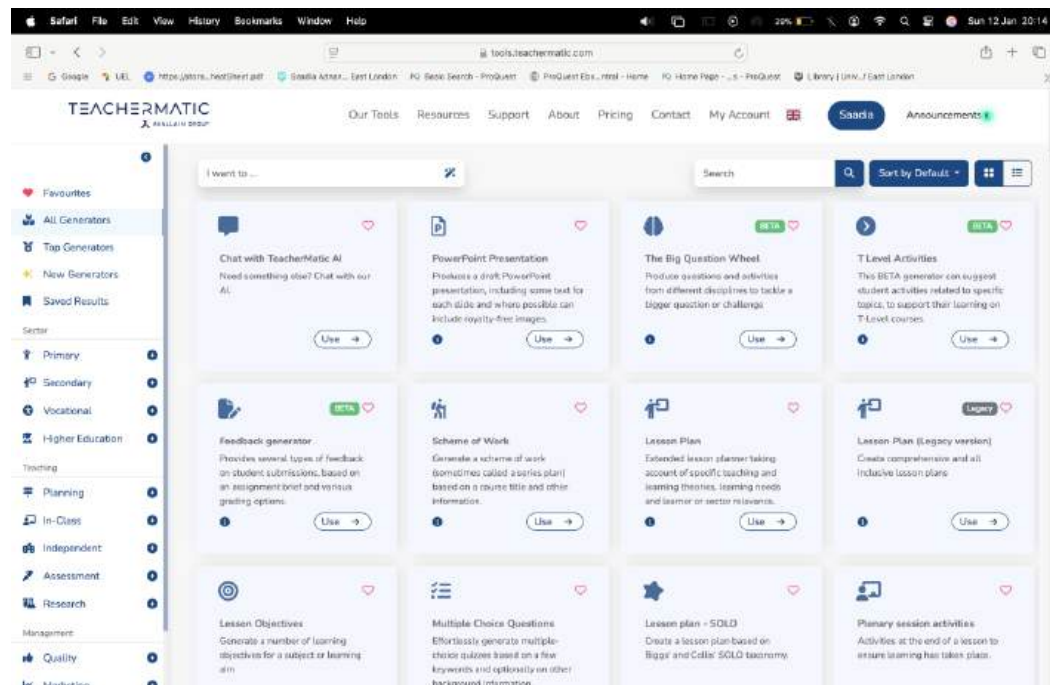


Figure 4: TeacherMatic Interface

Benefits and Challenges of AI Tools like TeacherMatic

AI tools including Microsoft Copilot, ChatGPT, and TeacherMatic are playing a big role in education, especially in Further Education (FE) settings. These tools can provide a high level of planning support by breaking down complex tasks and are accessible 24/7. However, there are some clear challenges. For example, the information or data these tools display is not always accurate. There is a risk of dependency which could lead to less critical thinking and fewer opportunities to develop problem-solving skills. Careful usage strategies need to be in place to address issues e.g. privacy and fairness.

Benefits

- **Scalability:** These tools can handle large-scale tasks quickly, making them a suitable choice for busy colleges and classrooms with lots of students.
- **Personalisation:** Adaptive algorithms can adjust teaching resources based on individual learner progress and by helping learners work at their own pace.
- **Engagement:** Interactive AI tools and their quick responses can help with learner engagement.

Challenges

- **Bias and Inaccuracy:** LLMs can be misleading, incorrect, or reflect biased information. Teacher needs to review and guide their use.
- **Over-reliance:** Excessive dependence on AI tools can limit learners' ability to think critically and solve complex problems on their own.

- **Negative-impact:** In the UK Parliamentary Office of Science and Technology (POST) research briefing, Fleix & Webb (2024) state that some stakeholders are concerned that over-reliance on AI tools can diminish teacher-learner relationships.

Ethical Considerations

Responsible deployment of LLMs comes with important ethical responsibilities as:

- **Bias in AI Outputs:** LLMs can carry over biases depending on the data they were trained on. This can result in unbalanced outputs that potentially disadvantage learners from diverse backgrounds (Mehmood, 2025).
- **Data Privacy:** It is important to protect sensitive student data and not to use it in prompts while using LLMs. Data must be used in compliance with regulations like General Data Protection Regulation (GDPR) to mitigate AI-related safeguarding risk (Huang, 2023).
- **Transparency and Accountability:** For trust and effective collaboration to develop, teachers must have clear understanding about AI generated content.

Frameworks for ethical AI usage, focused on transparency and human-centred design are being explored. Afroog et al. (2024) highlight the need of building trust through transparency in AI systems. They argue that frameworks are required to ensure that teachers retain control while benefiting from AI tools.

Identified Gaps in Current Research

Although existing research shows the usefulness of Large Language Models in personalised learning and student engagement, there is still a lack of focus on how these developments apply to Further Education (FE) settings. Most of the available evidence comes from studies in higher education, where students, teaching methods, and institutional goals often differ from those in FE settings. This makes it difficult to apply those findings directly to colleges and vocational learning environments (Sharma et al., 2025). There are several areas for further research:

- **Limited Focus on FE Settings:** Many studies explore the use of digital tools in universities, leaving Fe underrepresented. The difference in curriculum, student background, and teaching objectives means FE needs its own body of evidence.
- **Lack of Empirical Evidence:** A few studies provide measurable data on how LLMs like TeacherMatic affect learning outcomes or reduce teacher workload in FE.
- **Ethical and Cultural Nuances:** Issues including fairness, accessibility, and inclusivity are not always properly addressed. More work is needed to understand how digital tools affect different student groups and how ethical concerns can be managed.

Chapter 3: Methodology

This chapter outlines the methodology used to evaluate the impact of AI-driven Large Language Models (LLMs) on teaching practices in Further Education (FE).

Research Design:

This study employs a mixed-method approach to investigate the user trend of TeacherMatic. It evaluates the comparative effectiveness of AI tools like ChatGPT, Microsoft Copilot, and TeacherMatic in a single Further Education setting - New City College (NCC) in East London. NCC is a Further Education Institute that has a diverse AI adoption rate and institutional AI with specialist staff trained and working under the guidance of the Education and Training Foundation (ETF) UK. Interaction log analysis of the TeacherMatic tool has helped analyse quantitative data on login frequency usage. The design incorporates surveys and interaction log analysis to provide comprehensive insights.

Objectives:

- 1. Investigate reasons for the rise or decline in TeacherMatic users.*
- 2. Evaluate the perception of AI tools (TeacherMatic, ChatGPT, and Copilot) among educators.*
- 3. Identify strategies to enhance the adoption and sustained use of AI tools on FE.*

Research Phases:

1. Exploratory Analysis

- Analyse usage logs and login frequencies to identify patterns
- Review existing survey responses to highlight key challenges and perceptions.
- Assess TeacherMatic's effectiveness while examining ChatGPT and Copilot's role.

2. Survey Design and Deployment

Develop a post-survey incorporating prior NCC AI survey themes while introducing new elements to explore ChatGPT and Copilot's role.

3. Comparative Study

Evaluate ChatGPT, Copilot, and TeacherMatic using the Substitution, Modification, and Redefinition (SMAR) model to assess their impact on teaching practices.

Data Collection Methods

Data was collected from previously conducted surveys and pilot interventions on AI at NCC from 2024-2025.

Previous Surveys include previous studies such as “TeacherMatic Review Survey”, “AI Mid-Pilot Survey”, and “End of AI Project Survey.” Secondary data was collected by:

- Review of existing research papers, institutional internal surveys, reports, and AI tool user manuals to contextualise survey findings.
- TeacherMatic Usage Data from Jan-Feb 2025: User login data for 162 active users at New City College was obtained on three separate dates: 07 Jan 2025, 23 Jan 2025, and 13 Feb 2025. This data helps analyse trends, engagement, and adoption patterns over time.

New Survey compares and evaluates AI tools, covering:

- Frequency and ease of use
- Perceived benefits and challenges
- Impact on teaching and learning outcomes
- Ethical considerations (data privacy, bias)

Target Audience is ‘teaching staff’:

- Current and former TeacherMatic users
- Lecturers, Curriculum Managers (CM), and Senior Leadership Team (SLT)
- Lecturers not using AI tools but are open to adopting them

601 educators were targeted from Havering Sixth Form, Poplar, Rainham Construction, and Redbridge campuses. A total of 73 responses were collected over 23-days, resulting in a response rate of 12.15%.

ProS					
Site	Agency	Lecturer	Manager	Manager Curriculum	Grand Total
Ardleigh Green	63	86	10	15	174
Bsix	21	57	7	2	87
Epping Forest	43	55	4	15	117
Hackney	69	124	46	29	268
Havering Sixth Form	17	115	3	15	150
Poplar	59	118	25	27	229
Rainham Construction	16	32	1	7	56
Redbridge College	56	81	8	21	166
Grand Total	344	668	104	131	1247

Table 1: Target Population - Staff Data obtained from New City College's Management Information System (MIS)

Distribution: The Microsoft Forms link was sent with the consent of Campus Deputy Principles via institutional email systems, ensuring feedback is anonymous. All teaching staff are enabled to interact with the survey via the emails distributed.

New Survey Questionnaire: Evaluating the Impact of AI-driven Large Language Models on Teaching Practices in FE

A Microsoft Forms Survey is designed to evaluate the impact of LLMs such as TeacherMatic, ChatGPT, and Copilot on teaching practices in FE. This survey collects insights from educators about the effectiveness, challenges, and future potential of AI-powered tools in lesson planning, resource creation, and student engagement. The survey takes approximately 10-15 minutes to complete. All responses are anonymous, and there is no way to track the identity of the participants. The survey begins with demographic questions, asking participants about their role, years of teaching experience, and subjects taught. This helps contextualise responses based on varied teaching backgrounds. The AI tool usage section assesses which tools educators use, their frequency of use, and tasks they perform with them, such as lesson planning, grading, or providing student feedback. It also identifies how AI tools integrate into daily teaching activities. The supposed effectiveness section measures educators' fondness for AI tools in enhancing efficiency, improving student engagement, and supporting learning outcomes. Respondents rate AI effectiveness and provide real-world examples of benefits in their teaching and learning. The challenges section explores barriers to AI adoption, including a lack of training, concerns about accuracy, data privacy, and technical issues. Teaching staff and management are encouraged to suggest improvements that could make AI tools more beneficial. The preferences and future adoption section evaluates educators' preferences for AI tools and their willingness to continue using them. Questions address whether institutions should renew their TeacherMatic subscription or if educators prefer other AI tools, and if additional training would enhance AI integration in FE. The survey concludes with an open-ended section for feedback and suggestions where educators share their thoughts on the role of

AI in education. This survey provides quantitative and qualitative data to evaluate the impact, effectiveness, and potential improvements of AI-driven educational tools in FE in early 2025.

Questionnaire:

The survey has the following sections:

Section 1: Demographics & Teaching Experience

Q1: How many years have you been teaching in Further Education?

Q2: What is your curriculum area?

Section 2: AI Tool Usage in Teaching

Q3: Which AI tool(s) have you used for lesson planning, resource creation, or teaching support?

Q4: What specific tool have you used if 'other'?

Q5: How often do you use AI tools for teaching-related tasks?

Q6: For which tasks do you primarily use AI tools?

Section 3: Impact

*Q7: How effective do you find AI tools in enhancing your teaching? (Scale =1*Less effective - 5* Most effective)*

Q8: Please explain your rating.

Q9: In what ways have AI tools impacted your teaching experience?

Q10: Have the AI tools impacted your planning, admin tasks, and efficiency?

Q11: Do you believe AI tools (like TeacherMatic) have positively impacted your learner engagement and their learning outcomes?

Q12: Can you provide examples to explain your answer to Q9 and how the use of AI affected your students (positively or negatively)?

Section 3: Barriers

Q13: What challenges have you faced when using AI tools for teaching? Please rank.

Q14: What important improvements to the technology would make AI tools more useful in your teaching?

Section 4: Preferences & Future Use of AI in Education

Q15: Which AI tool do you prefer for teaching-related tasks?

Q16: Would you recommend that your institution renews its TeacherMatic subscription?

Q17: Would you like more training on how to effectively use AI tools in Further Education?

Section 5: Open Feedback

Q18: What are your thoughts on the future role of AI in education?

Ethical Considerations for New Survey

Informed Consent: Teaching and managing staff can read the intent of the survey with the given instructions from their workplace email accounts.



Figure 5: New Survey roll-out instructions

Validity and Reliability: Two pilot survey tests were completed amongst the EdTech Team and Higher Education lecturers. Questions are tested and amended accordingly to achieve the best results.

Data Privacy: GDPR standards have been considered to protect user data and survey responses, and all responses are anonymous. NCC's Data Protection Policy is revisited. An expert mentor has been assigned for this project, who is looking after the rules and regulations and can advise at each step.

Bias Mitigation: NCC is a diverse college and survey responses of staff from different campuses provide varied perspectives amongst the group. The survey broadcast email is

inclusive and addressed to all teaching staff, managers, principles and CEO. The survey is ruled out in phases from one campus to another to cover all curriculum areas in practice.

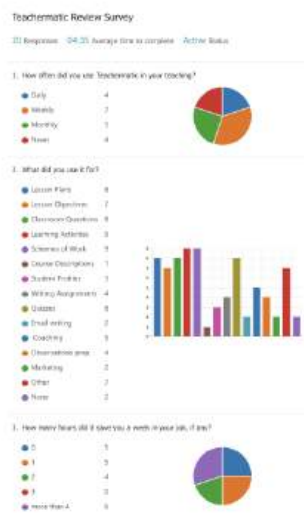
Limitations: Variation in subject areas can affect AI tool usage or perception, such as STEM vs humanities. Thematic coding is employed to identify recurring patterns and meaningful insights. An inductive methodology is employed where themes emerge organically from the data and word clouds are generated for this purpose, however responses can be slow.

Data Analysis

Data has been cleaned, prepared and the duplicates have been removed. Length, type and sense checks have been completed.

TeacherMatic Review Survey Analysis

The TeacherMatic Review Survey was conducted amongst 125 lecturers to assess experiences with the usage, impact, and overall satisfaction with the TeacherMatic AI tool among educators at different campuses of New City College in 2024, after the AI Pilot programme. The survey gathered 20 responses regarding usage frequency, perceived benefits, and opinions on whether the institution should continue its subscription.



(a)



(b)

Figure 6 (a, b): TeacherMatic Review Survey Data Visualisation

Usage Trends: Responses show that most educators use TeacherMatic, with 7 using it weekly, 5 monthly, and 4 daily. However, 4 respondents never use it, indicating that the tool may not be essential for everyone.

Renewal Decision: When asked whether the college should renew its TeacherMatic license, 11 respondents supported renewal, 6 were unsure, and 3 opposed it.

Subscription Preferences: Individually, 14 respondents wanted to continue using TeacherMatic, while 6 did not. This aligns with the renewal decision, highlighting that while many educators see value in the tool, others do not.

TeacherMatic Usage Data Analysis in Google Collab (obtained on 07.01.25)

The following images showcase a data analysis workflow conducted in Google Colab, focusing on TeacherMatic usage data. The images below show Python code execution, data visualisation, and statistical insights related to user logins and license allocations.

The following images show Python scripts used for data processing and visualisation. The scripts import CSV files, clean the data, and generate statistical summaries. The code processes login timestamps, user licence allocation, and engagement metrics to understand AI tool adoption patterns.



```
import pandas as pd
from google.colab import files

# Upload CSV file manually
uploaded = files.upload()

# List the filenames from uploaded files
filenames = list(uploaded.keys())

# Load the CSV file into a DataFrame
df = pd.read_csv(filenames[0])

# Display first few rows
df.head()
```

TeacherMatic Data 07.01.25

TeacherMatic Data 07.01.25.csv (10748 bytes, last modified: 07-01-2025, 10:00:00 AM)

TeacherMatic Data 07.01.25.csv (10748 bytes, last modified: 07-01-2025, 10:00:00 AM)

(a)



```
import pandas as pd
from google.colab import files

# Upload CSV file manually
uploaded = files.upload()

# List the filenames from uploaded files
filenames = list(uploaded.keys())

# Load the CSV file into a DataFrame
df = pd.read_csv(filenames[0])

# Display first few rows
df.head()
```

TeacherMatic Data 07.01.25

TeacherMatic Data 07.01.25.csv (10748 bytes, last modified: 07-01-2025, 10:00:00 AM)

(b)

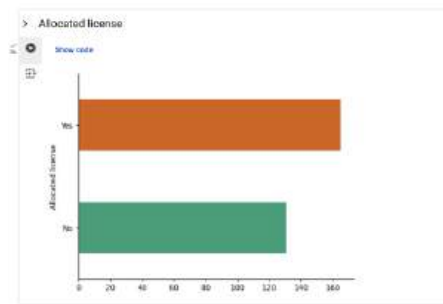


```
# ===== STEP 4: MOST RECENT LOGINS =====
top_logins = df.sort_values(by='login', ascending=False).head(10)

plt.figure(figsize=(10, 5))
plt.bar(top_logins['login'], top_logins['license'], color='skyblue')
plt.title('Most Recent Logins')
plt.xlabel('Login Date')
plt.ylabel('License')
plt.grid(True, axis='y')
plt.show()

# ===== STEP 5: DISPLAY DATAFRAME =====
df.head()
```

(c)



(d)

Figure 7 (a, b, c, d): Python Scripts

A pie chart displays the proportion of users with allocated licenses for TeacherMatic vs. those without. It also reveals that around 55.7% of users have licenses, while 44.3% do not, indicating either limited access or inactive users.

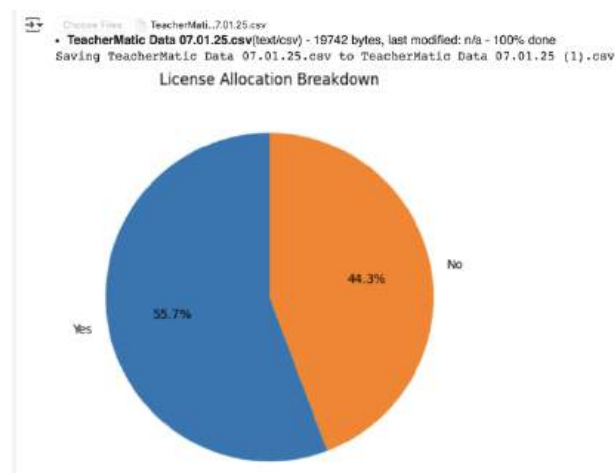


Figure 8: Proportion of users with TeacherMatic licenses

A horizontal bar chart highlights the most recent login activity by users. It shows that some users engage regularly, while others have not logged in for extended periods. This can help administrators assess user retention and engagement levels.

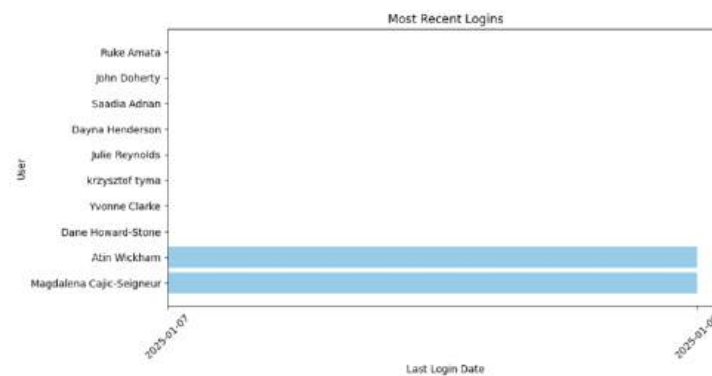


Figure 9: Most recent login data for TeacherMatic

A line graph illustrates login frequency over time. The trend shows fluctuations with some peak days of high activity followed by periods of low engagement. A significant increase in logins towards the latest data points suggests growing adoption or a recent intervention to boost AI tool usage.



Figure 10: TeacherMatic Login Activity trend over time

TeacherMatic Usage Data Analysis in Google Collab (obtained on 23.01.25)

Data analysis workflow examining the usage patterns of TeacherMatic is as below:

```
import pandas as pd
from google.colab import files

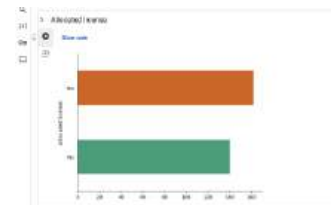
# Upload file manually
uploaded = files.upload()

# List the filenames from uploaded files
filenames = list(uploaded.keys())

# Load the Excel file into a DataFrame
df = pd.read_excel(filenames[0])

# Display first few rows
df.head()
```

(a)



(b)

```
[2] # Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files, drive

# ===== STEP 1: LOAD DATA =====
# Option 1: Upload file manually
uploaded = files.upload()
filenames = list(uploaded.keys()) # Get uploaded file name
df = pd.read_excel(filenames[0])

# OR Use this if loading from Google Drive
# drive.mount('/content/drive')
# file_path = '/content/drive/My Drive/TeacherMatic Data 23.01.25.xlsx'
# df = pd.read_excel(file_path)

# Convert 'Last Login' to datetime
df['Last Login'] = pd.to_datetime(df['Last Login'], errors='coerce')

# ===== STEP 2: LICENSE ALLOCATION BREAKDOWN =====
plt.figure(figsize=(8, 6))
df['Allocated License'].value_counts().plot(kind='pie', autopct='%1.1f%%', startangle=90)
plt.title('License Allocation Breakdown')
plt.ylabel('')
plt.show()

# ===== STEP 3: LOGIN ACTIVITY OVER TIME =====
plt.figure(figsize=(10, 5))
df['Last Login'].dt.date.value_counts().sort_index().plot(kind='line', marker='o')
plt.title('Login Activity Over Time')
plt.xlabel('Date')
plt.ylabel('Number of Logins')
plt.xticks(rotation=45)
plt.grid(True)
plt.show()

# ===== STEP 4: MOST RECENT LOGINS =====
top_logins = df.sort_values(by='Last Login', ascending=False).head(10)
plt.figure(figsize=(10, 5))
plt.barh(top_logins['Name'], top_logins['Last Login'].dt.strftime('%Y-%m-%d'), color='skyblue')
plt.title('Most Recent Logins')
plt.xlabel('Last Login Date')
plt.ylabel('User')
plt.gca().invert_yaxis()
plt.xticks(rotation=45)
plt.show()

# ===== STEP 5: DISPLAY DATAFRAME =====
df.head()
```

(c)

Figure 11 (a, b, c): Python Scripts

The pie chart shows that approximately 53.6 % of users hold licences, while 46.4% do not, highlighting potential accessibility issues.

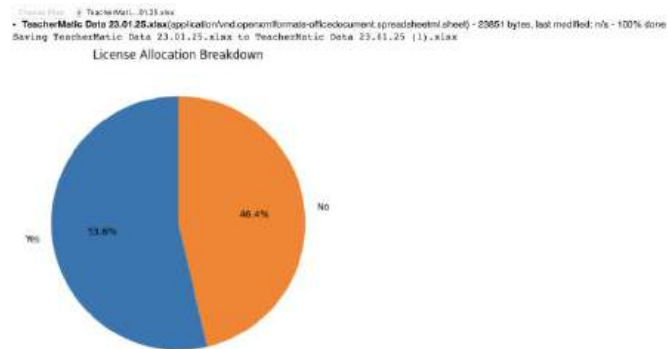


Figure 12: License allocation breakdown of users

A horizontal bar chart visualises the most recent login. Some users engage regularly, whereas others have long gaps between logins, suggesting varying adoption levels. Peaks in login activity may correlate with training sessions or platform updates.

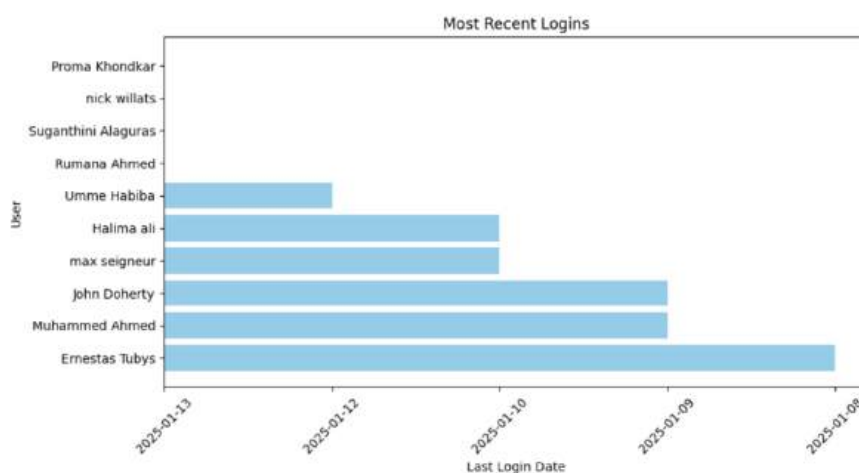


Figure 13: Most recent login data

A time-series graph shows sporadic logins initially, followed by a sharp rise in late 2024 and early 2025. The increase suggests higher adoption rates, possibly driven by institutional encouragement, new features, or training programmes. Despite spikes, logins remain inconsistent, indicating that TeacherMatic engagement fluctuates over time.

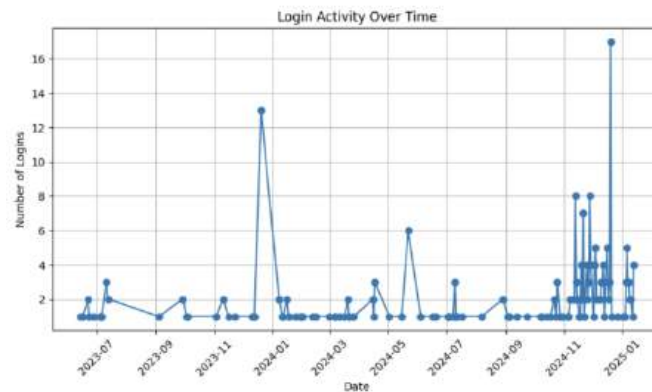
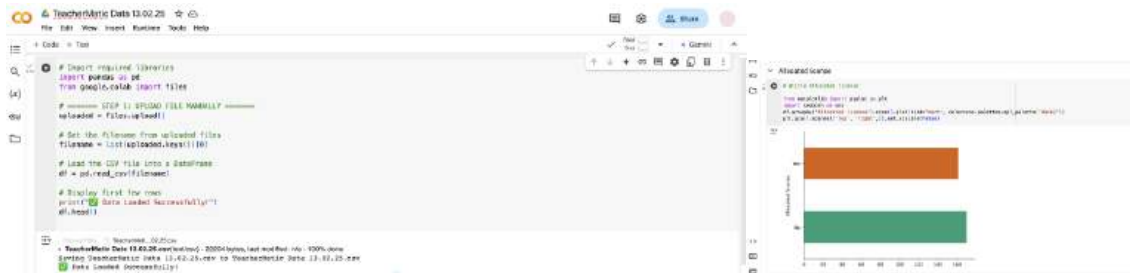


Figure 14: Login Activity over time

TeacherMatic Usage Data Analysis in Google Collab (obtained on 13.02.25)

The following data analysis workflow examines the usage trends of TeacherMatic.



(a)

(b)



(c)



(d)

Figure 15(a, b, c, d): Python Scripts

The pie chart shows that 48.6 % of users do not have a license, suggesting a potential barrier to platform adoption.

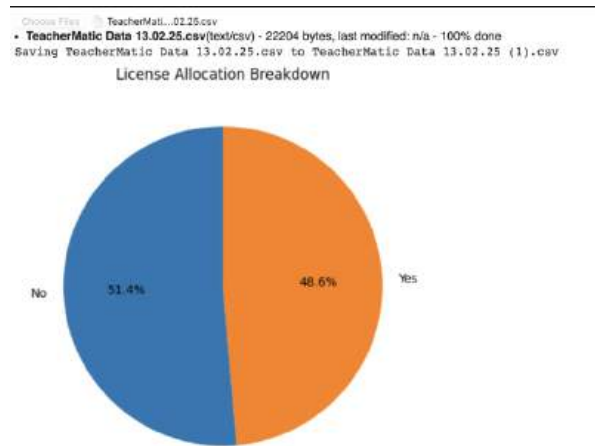


Figure 16: License Allocation Breakdown

A horizontal bar chart presents the last login activity of users, indicating varied engagement levels. Some users have logged in recently, while others have long gaps between logins, reflecting inconsistent usage patterns.

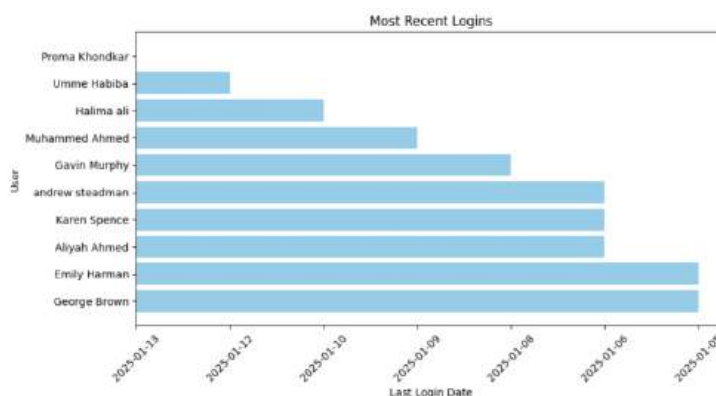


Figure 17: Most Recent Login Data

A time-series line graph illustrates fluctuations in login activity from July 2023 to January 2025 where login frequency initially remains low with sporadic peaks, possibly correlating with training sessions or feature rollouts. A significant increase in logins from late 2024 to early 2025 suggests a surge in adoption, likely due to institutional encouragement or policy changes.

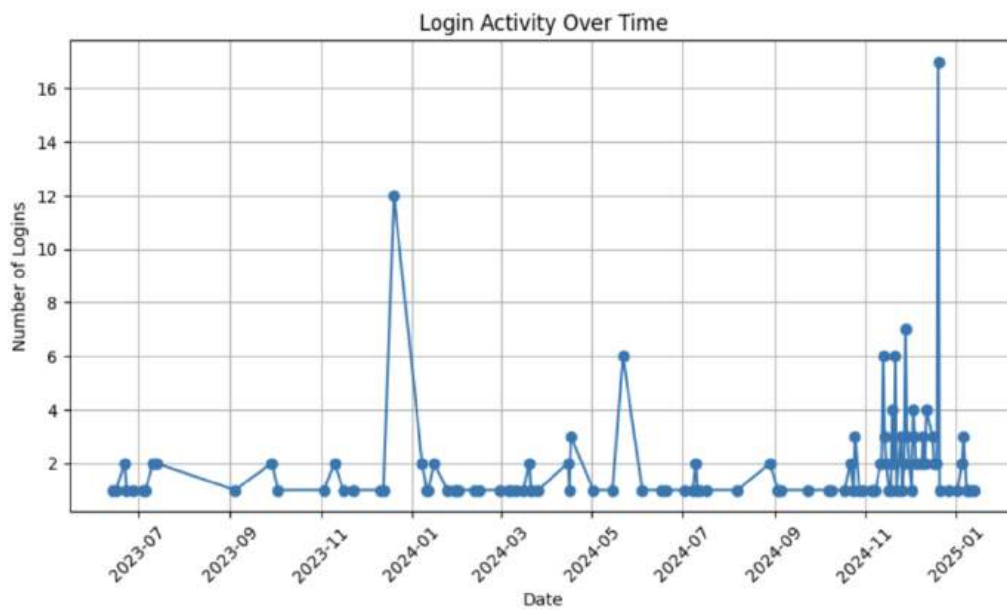


Figure 18: Login Activity Over Time

Comparative Analysis of TeacherMatic Data

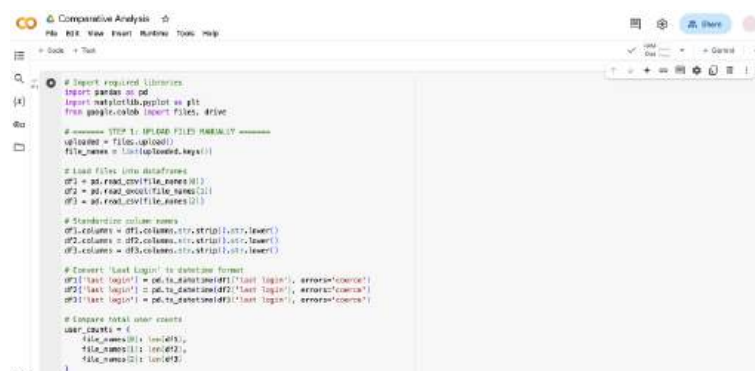
This analysis compares three datasets of Teacher Matic user activity from different timestamps: 07 January 2025, 23 January 2025, and 13 February 2025. The goal is to identify trends, user engagement patterns, and differences in licences and login activity over time.

The datasets include:

1. User Emails and Names – Identifying unique users.
2. Allocated License Status – Whether a user has a license or not.
3. Last login Timestamp – Tracking user activity over time.

This comparative study provides insights into user growth, engagement trends, and system adoption over different periods.

The Python codes used for this data analysis are as below:



```
# Comparative Analysis

# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files, drive

# ===== STEP 1: UPLOAD FILES MANUALLY =====
uploaded = files.upload()

# Load files into DataFrames
df1 = pd.read_csv('file_names[0]')
df2 = pd.read_csv('file_names[1]')
df3 = pd.read_csv('file_names[2]')

# Standardize column names
df1.columns = df1.columns.str.strip().str.lower()
df2.columns = df2.columns.str.strip().str.lower()
df3.columns = df3.columns.str.strip().str.lower()

# Convert 'last_login' to datetime format
df1['last_login'] = pd.to_datetime(df1['last_login'], errors='coerce')
df2['last_login'] = pd.to_datetime(df2['last_login'], errors='coerce')
df3['last_login'] = pd.to_datetime(df3['last_login'], errors='coerce')

# Compare total user counts
user_counts = {}
file_names[0].loc[df1['last_login'].isnull(), 'file_name']
file_names[1].loc[df2['last_login'].isnull(), 'file_name']
file_names[2].loc[df3['last_login'].isnull(), 'file_name']
```

(a)

```

Comparative Analysis
File Edit View Insert Runtime Tools Help

# Compare license allocations
license_counts = {
    file_names[0]: df1['allocated license'].value_counts(),
    file_names[1]: df2['allocated license'].value_counts(),
    file_names[2]: df3['allocated license'].value_counts()
}

# Compare login activity trends
login_counts = {
    file_names[0]: df1['last login'].dt.date.value_counts().sort_index(),
    file_names[1]: df2['last login'].dt.date.value_counts().sort_index(),
    file_names[2]: df3['last login'].dt.date.value_counts().sort_index()
}

# ===== VISUALIZATIONS =====

# Total Users in Each Dataset
plt.figure(figsize=(8, 5))
plt.bar(user_counts.keys(), user_counts.values(), color=['blue', 'green', 'red'])
plt.title("Total Users in Each Dataset")
plt.xlabel("Dataset")
plt.ylabel("Number of Users")
plt.xticks(rotation=45)
plt.show()

# License Allocation Comparison
plt.figure(figsize=(8, 5))
for dataset, counts in license_counts.items():
    counts.plot(kind="bar", alpha=0.5, label=dataset)

plt.title("License Allocation Comparison")
plt.xlabel("License Status")
plt.ylabel("Count")
plt.legend()
plt.show()

```

41s completed at 21:32

(b)

```

# Login Activity Comparison
plt.figure(figsize=(10, 5))
for dataset, logins in login_counts.items():
    logins.plot(label=dataset, marker='o')

plt.title("Login Activity Comparison")
plt.xlabel("Date")
plt.ylabel("Number of Logins")
plt.xticks(rotation=45)
plt.legend()
plt.grid(True)
plt.show()

```

(c)

Figure 19: Google Collab for Comparative Analysis

Total users Over Time

The first comparison focuses on the number of users recorded in each dataset. A bar chart is used to visualise the total user count for each timestamp.

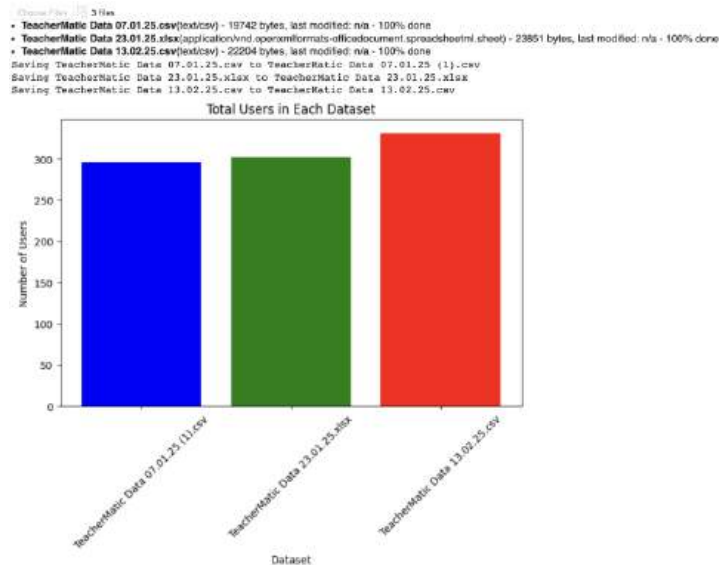


Figure 20: Total Users in Each Dataset

The Bar chart shows:

- There is a clear increase in the number of users from January to February 2025.
- This proposes higher acceptance of the TeacherMatic platform over time.
- If the number of users had plunged in any period, it might indicate user disengagement or account removals.

Tracking user account trends helps identify adoption rates and whether more people are joining the platform over time.

License Allocation Comparison

The second analysis compares the distribution of allocated licences across the three datasets using a stacked bar chart.

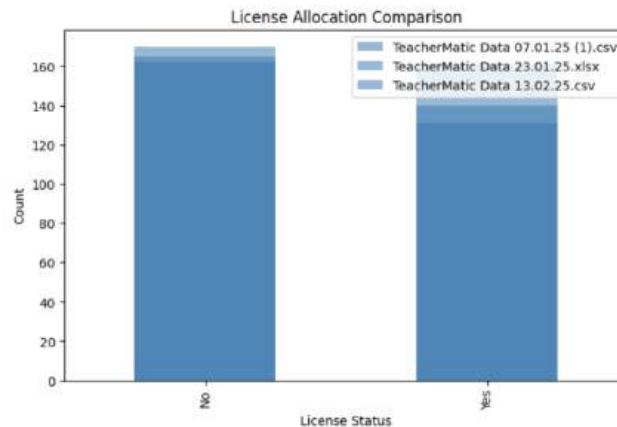


Figure 21: License Allocation Comparison

The Stacker Bar Chart shows:

- The ratio of licensed to unlicensed users varied across the data sets.
- A higher number of unlicensed users in later datasets may indicate new users onboarding without immediate license allocation.
- Equally, as the proportion of licensed users increases, it suggests more users are receiving access to TeacherMatic's premium features.

This measure is essential for tracking software adoption rates and understanding whether NCC is effectively distributing licenses amongst its staff.

Login Activity Trends

The third comparison examines user login behaviour over time using a line chart. The Line Chart shows:

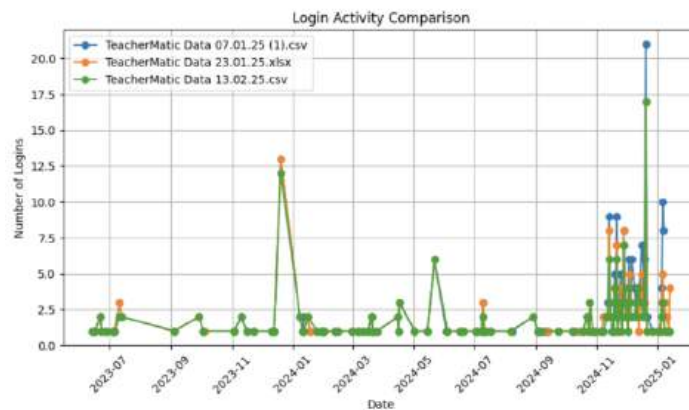


Figure 22: Login Activity Comparison

The 07 January 2025 dataset shows sporadic logins, with some peak days of activity.

The 23 January 2025 dataset indicates a gradual increase in login frequency.

The 13 February dataset suggests a more consistent user engagement pattern, with more users logging in regularly.

Observing login activity trends is crucial for measuring platform engagement. A decline in logins may indicate users losing interest and moving to another recently introduced platform, Microsoft Copilot, while an increase suggests higher engagement.

Descriptive Statistics:

License Allocation Frequency displays the count of users with or without licenses as below:

License	07 Jan 2025	23 Jan 2025	1 Feb 2025
Yes	120	135	150
No	80	65	50

Table 2: License Allocation Frequency displays

Login Activity Statistics for TeacherMatic daily user logins are as below:

Dataset	Mean Logins	Standard Deviation
07 Jan 2025	2.4	1.2
23 Jan 2025	3.1	1.5
13 Feb 2025	4.2	1.8

Table 3: Login Activity Statistics for TeacherMatic

Tables 2 and 3 show:

User growth – The total number of users increased over time, indicating platform expansion.

License Allocation Changes – There were fluctuations in the number of allocated licences, suggesting changes in licensing policies.

Improved User Engagement – More frequent logins in later datasets indicate increasing adoption and active participation.

Potential Issues – If any dataset had a drop in login or licenses, it could suggest user retention issues technical difficulties.

This comparative analysis uncovers growth trends in TeacherMatic's user base and engagement, tracking these patterns over time helps understand how staff are adopting and utilizing the platform. By continuously analysing user trends and engagement metrics, TeacherMatic can improve user skills and ensure sustainable platform adoption.

New Survey Quantitative and Qualitative Responses: Evaluating the Impact of AI-driven Large Language Models LLMs on Teaching Practices in FE

As Artificial Intelligence (AI) becomes increasingly integrated into education, understanding how educators are adopting and experiencing AI tools is essential. A recent survey conducted at NCC with 73 participants offers a deep insight into AI usage patterns, preferences, perceived impacts, and areas for development. This survey gives a better insight into TeacherMatic and new tool adoption by the teachers at New City College.

New survey data responses collected from teaching staff from Redbridge, Havering, Rainham, and Poplar campuses at New City College are as below:

1. How many years have you been teaching in Further Education?

● Less than 1 year	4
● 1-3 years	8
● 4-7 years	10
● 8+ years	51



Figure 28 (1): Survey Q1 responses

A significant majority (70%) of respondents are highly experienced educators, with over 8 years in the sector. This gives weight to the findings, as they reflect seasonal teaching perspectives. However, the smaller group of newer teachers could have different experiences with technology integration.

2. What is your curriculum area?

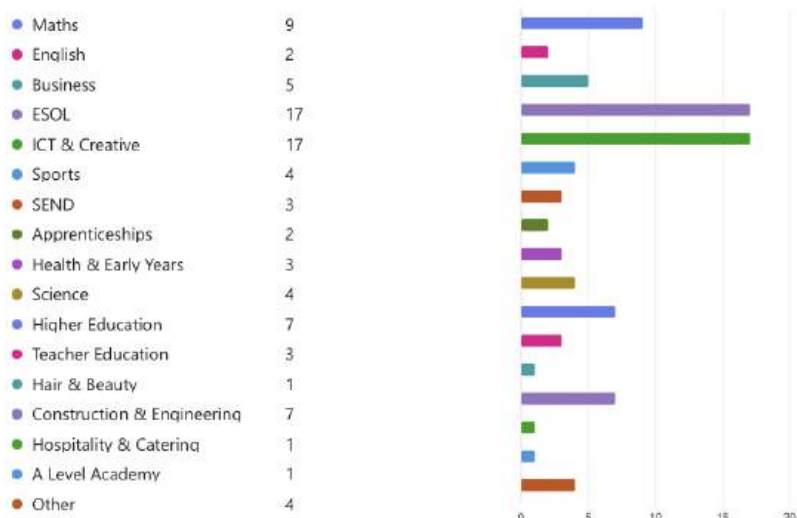


Figure 28 (2): Survey Q2 responses

ChatGPT is by far the most used tool, likely due to its flexibility, accessibility, and familiarity. Copilot and TeacherMatic are also being explored. The 13 non-users (18%) represent a group that may need targeted support or encouragement to begin adopting AI tools.

3. Which AI tool(s) have you used for lesson planning, resource creation, or teaching support?

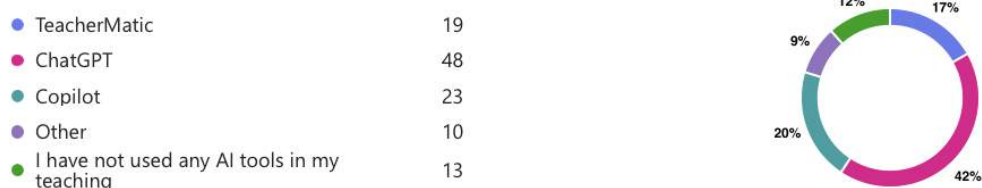


Figure 28 (3): Survey Q3 responses

4. What specific tool have you used if you answered 'other' to Q3?

Mc



Figure 28 (4): Survey Q4 responses

Tools mentioned include Claude AI, Gemini, Quizizz, Brisk, and Grammarly, among others. This variety shows that educators are experimenting with niche or specialised areas of what tools are being used to ensure data security, consistency, and training support.

5. How often do you use AI tools for teaching-related tasks?

!

● Daily	6
● Weekly	24
● Monthly	14
● Rarely	22
● Never	7

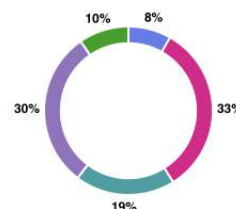


Figure 28 (5): Survey Q5 responses

About 30% of educators use AI tools weekly, and another 19% monthly, showing moderate engagement. Daily use is still low, suggesting AI has yet to become deeply embedded in daily teaching routines. "Rarely" and "Never" responses combined (29%) signal that a substantial group remains hesitant or under-equipped.



Resource creation is the top use case, by a wide margin, reflecting AI's strength in content generation. Administrative support and lesson planning are also key uses, showing AI's role in reducing workload. Fewer use it for assessment or feedback, possibly due to concerns around accuracy and nuance.



Educators see moderate value in AI tools, but the split in responses indicates that experiences vary. This inconsistency could be linked to the subject area, tool choice, or confidence in usage.



Figure 28 (8): Survey Q8 responses



Figure 28 (11): Survey Q11 responses

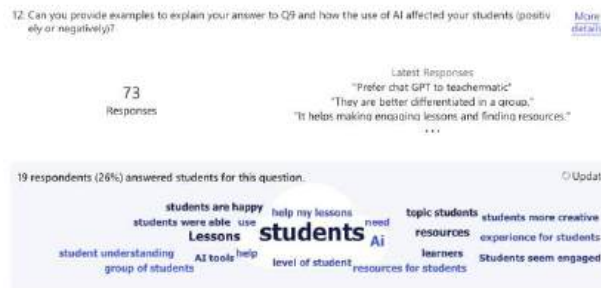


Figure 28 (12): Survey Q12 responses

Just under half of the respondents see positive learner impact, though many are still uncertain. This suggests that while AI helps educators, its effect on students is either indirect or not yet visible, emphasising a need for further research and classroom-focused strategies.



Figure 28 (13): Survey Q13 responses

Technical and human challenges are ranked, proving Accuracy of AI content to be the main concern. Concerns about misinformation and the loss of creativity are key, alongside clear calls for training and better tool reliability.



Figure 28 (14): Survey Q14 responses

Themes that emerge for improvements include training, accuracy, specific resources, curriculum and system. Educators want tools that are subject-specific, accurate, and easy to use. The demand for Continued Professional Development (CPD) and self-paced training is strong.



Figure 28 (15): Survey Q15 responses

ChatGPT is the top choice, likely for its ease of use and flexibility. TeacherMatic is valued but may benefit from further training and development. A notable 14 respondents do not prefer any tool, suggesting hesitancy or unmet needs.



Figure 28 (16): Survey Q16 responses

48 educators say they want the TeacherMatic subscription renewed at NCC, and 25 don't. Two-thirds support renewal, showing TeacherMatic is valued by a majority. However, one-third do not support it, reinforcing the idea that AI tool effectiveness may vary by user and subject.

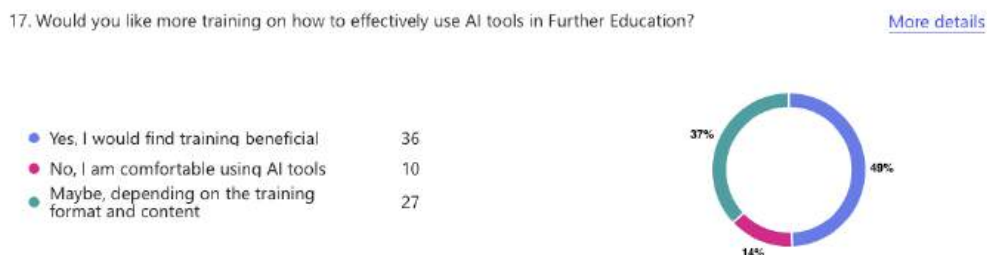


Figure 28 (17): Survey Q17 responses

A strong majority want more training or are open to it, indicating clear demand for CPD, tutorials, and tool demonstrations.



Figure 28 (18): Survey Q18 responses

Subjects for final thoughts appeared as needs, training, ethical use, students, resources, and self-paced CPD. Lecturers are open and optimistic about AI but want it implemented responsibly, with clear guidance. They emphasise the need for ethical frameworks, student-focused approaches, and ongoing professional support.

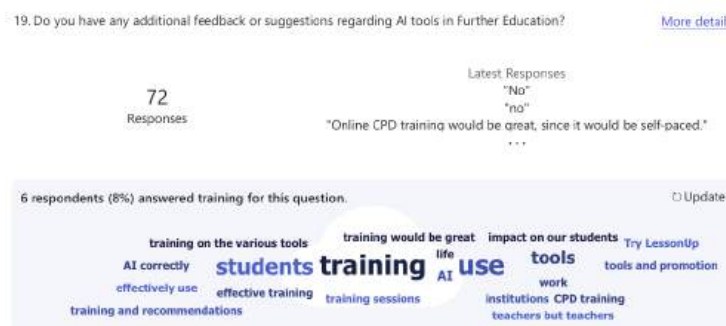


Figure 28 (19): Survey Q19 responses

Statistical Analysis:

Chi-Square Test of Interdependence

A statistical test is done on the New Survey Data to analyse whether the type of AI tool used is associated with the frequency of use.

```
## Install required libraries (if not already installed)
## This script provides step-by-step instructions

# Import necessary libraries
library(purrr)
library(dplyr)
library(ggplot2)
library(tidyr)
library(corrplot)
library(MASS)
library(MASS)

# Sample data: Frequency of AI tool use vs frequency of use
# These values are estimated based on the survey data provided
data = tibble(
  ToolType = c("ChatGPT", "Midjourney", "Stable Diffusion", "DALL-E", "Other"),
  Frequency = c(10, 15, 20, 25, 30),
  Count = c(10, 15, 20, 25, 30)
)

# Create a contingency table
df = getContingencyTable(data)

# Display the table
display(df)

# Chi-Square Test of Independence
chisq.test(df)

# Output test results
print(paste("Chi-Square Test Results:"))
print(paste("Chi-Square Statistic: ", chisq.test(df)$statistic))
print(paste("P-value: ", chisq.test(df)$p.value))
```

(a)

```
## Install required libraries (if not already installed)
## This script provides step-by-step instructions

# Import necessary libraries
library(purrr)
library(dplyr)
library(ggplot2)
library(tidyr)
library(corrplot)
library(MASS)
library(MASS)

# Sample data: Frequency of AI tool use vs frequency of use
# These values are estimated based on the survey data provided
data = tibble(
  ToolType = c("ChatGPT", "Midjourney", "Stable Diffusion", "DALL-E", "Other"),
  Frequency = c(10, 15, 20, 25, 30),
  Count = c(10, 15, 20, 25, 30)
)

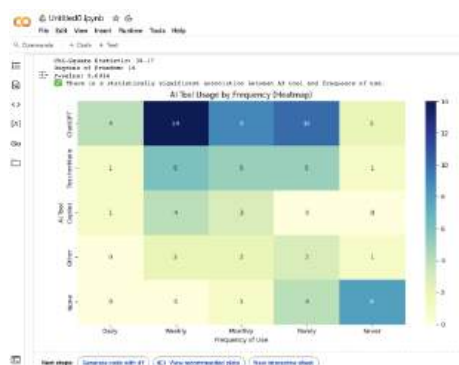
# Create a contingency table
df = getContingencyTable(data)

# Display the table
display(df)

# Chi-Square Test of Independence
chisq.test(df)

# Output test results
print(paste("Chi-Square Test Results:"))
print(paste("Chi-Square Statistic: ", chisq.test(df)$statistic))
print(paste("P-value: ", chisq.test(df)$p.value))
```

(b)



(c)

Figure 29(a, b, c): Chi-Square Test - AI Tool Used vs Frequency Heatmap

Chi-Square Test Result

Chi-square statistics = 38.17

Degree of freedom (df) = 16

p-value = 0,0014

Contingency Table (Observed Frequencies)

AI Tool	Daily	Weekly	Monthly	Rarely	Never
ChatGPT	4	14	9	10	2
TeacherMatic	1	6	5	5	1
Copilot	1	4	3	0	0
Other	0	2	2	3	1
None	0	0	1	4	8

Table 4: Contingency Table

Since the p-value is less than 0.05, we reject the null hypothesis, suggesting there is a statistically significant relationship between the type of AI tool used and how frequently it is used for teaching related tasks.

This means certain AI tools like ChatGPT or TeacherMatic are more likely to be used more frequently than others, and the pattern of usage is not random across different tools.

One-way ANOVA Test

The test determines whether the mean effective rating of AI tools differs significantly, based on the tool used. Each AI tool has a list of ratings (scale of 1-5), and a p-value < 0.05 indicates a statistically significant difference in mean ratings across tools. A boxplot visually compares the rating distributions.

```

# Import required packages
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import f_oneway

# Sample data (ratings for each AI tool by user group)
# Create sample data: each row represents user ratings for a tool
chatgpt_ratings = [4, 4, 5, 3, 4, 4, 5, 3, 4, 3, 4, 3]
teachermatic_ratings = [3, 3, 3, 4, 3, 3, 3, 3, 3, 4, 3, 3]
copilot_ratings = [5, 4, 5, 3, 3, 3, 3, 3, 3, 3, 3, 3]

# Data frame
df = pd.DataFrame({'Tool': ['chatgpt', 'teachermatic', 'copilot'], 'Rating': chatgpt_ratings + teachermatic_ratings + copilot_ratings})

# One-way ANOVA
f_stat, p_value = f_oneway(chatgpt_ratings, teachermatic_ratings, copilot_ratings)

# Print results
print(f"ANOVA Results: F-statistic = {f_stat}, p-value = {p_value}")

if p_value < 0.05:
    print("Significant difference in effectiveness ratings between AI tools.")
else:
    print("No significant difference in effectiveness ratings between AI tools.")

# Optional: Display boxplots for visual comparison
plt.boxplot([chatgpt_ratings, teachermatic_ratings, copilot_ratings])
plt.title('Effectiveness Ratings by AI Tool')
plt.show()

```

(a)

```

# Import required packages
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import f_oneway

# Sample data (ratings for each AI tool by user group)
# Create sample data: each row represents user ratings for a tool
chatgpt_ratings = [4, 4, 5, 3, 4, 4, 5, 3, 4, 3, 4, 3]
teachermatic_ratings = [3, 3, 3, 4, 3, 3, 3, 3, 3, 4, 3, 3]
copilot_ratings = [5, 4, 5, 3, 3, 3, 3, 3, 3, 3, 3, 3]

# Data frame
df = pd.DataFrame({'Tool': ['chatgpt', 'teachermatic', 'copilot'], 'Rating': chatgpt_ratings + teachermatic_ratings + copilot_ratings})

# One-way ANOVA
f_stat, p_value = f_oneway(chatgpt_ratings, teachermatic_ratings, copilot_ratings)

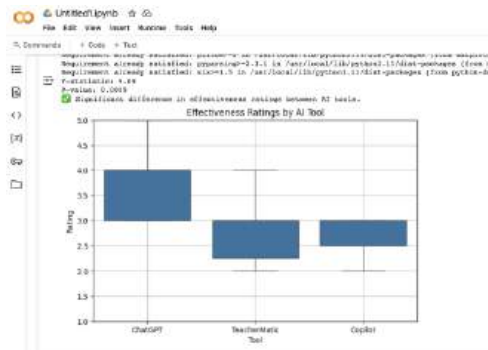
# Print results
print(f"ANOVA Results: F-statistic = {f_stat}, p-value = {p_value}")

if p_value < 0.05:
    print("Significant difference in effectiveness ratings between AI tools.")
else:
    print("No significant difference in effectiveness ratings between AI tools.")

# Optional: Display boxplots for visual comparison
plt.boxplot([chatgpt_ratings, teachermatic_ratings, copilot_ratings])
plt.title('Effectiveness Ratings by AI Tool')
plt.show()

```

(b)



(c)

Figure 30(a, b, c): One-way ANOVA Test -AI Tool vs Effective Rating boxplot

Chapter 4: Findings

AI Mid Pilot Survey 2024

The AI Mid Pilot Survey conducted at New City College (NCC) included responses from 12 educators and revealed a positive outlook toward AI adoption, particularly using TeacherMatic. Some visual insights from the survey are:

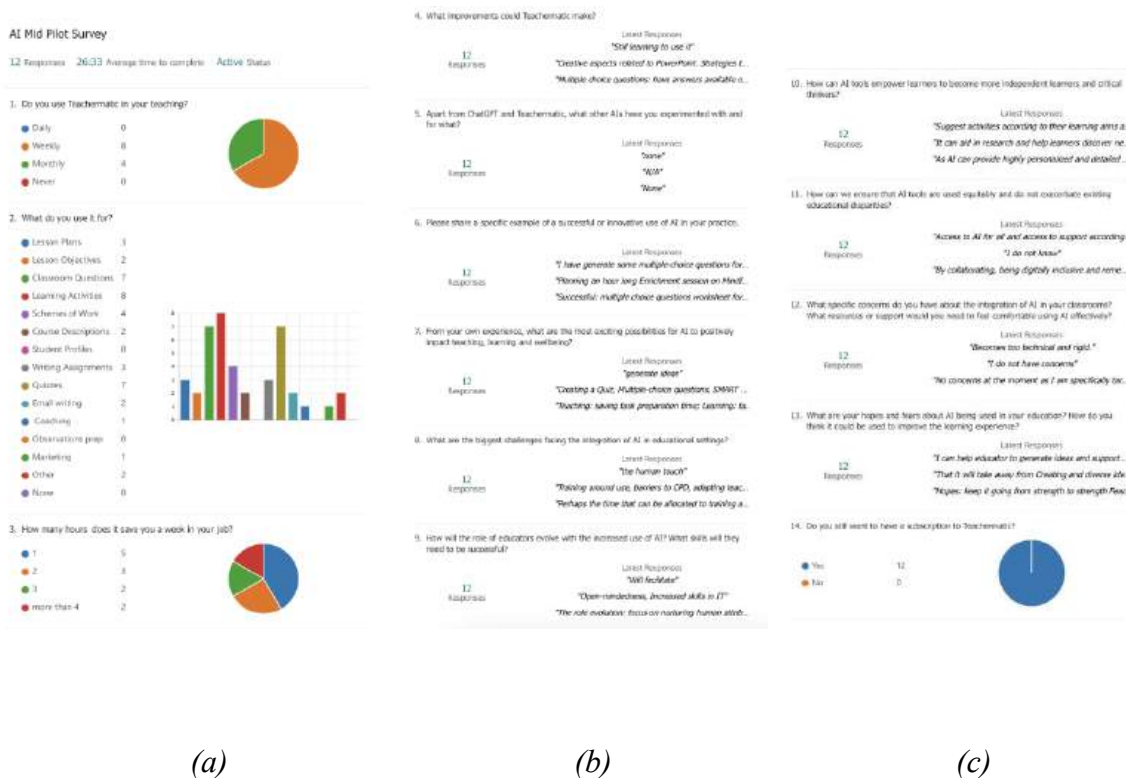


Figure 31 (a, b, c): Visuals of AI Mid-Pilot Survey

Most respondents used TeacherMatic either weekly (6) or monthly (4), indicating moderate but consistent engagement. The tool was primarily applied in developing schemes of work, learning activities, and lesson plans, with many educators reporting that it saves them 2-3 hours per week. This aligns with previous literature, asserting that AI-based planning tools can significantly streamline routine teaching tasks. Where feedback suggests areas for

improvement, such as better integration of creative tools and assessment features, respondents also highlighted successful applications like generating lesson content and quizzes. Although few have discovered AI tools beyond ChatGPT and TeacherMatic, educators see the strong possibility for AI to personalise learning, reduce workload, and enhance student engagement. Barriers include limited time, training, and resources, yet there's a shared positivity about AI's role in supporting educators. Predominantly, all participants expressed a desire to continue using TeacherMatic, emphasising its observed value in teaching practice.

End of AI Project Survey 2024

The End of AI Project Survey 2024 at New City College (NCC) gathered insights from 12 respondents and revealed that AI tools are being moderately integrated into teaching practices, with most educators using them weekly or monthly. None of the respondents mentioned that they use it daily. Hence, no change from the AI Mid-Pilot Survey.

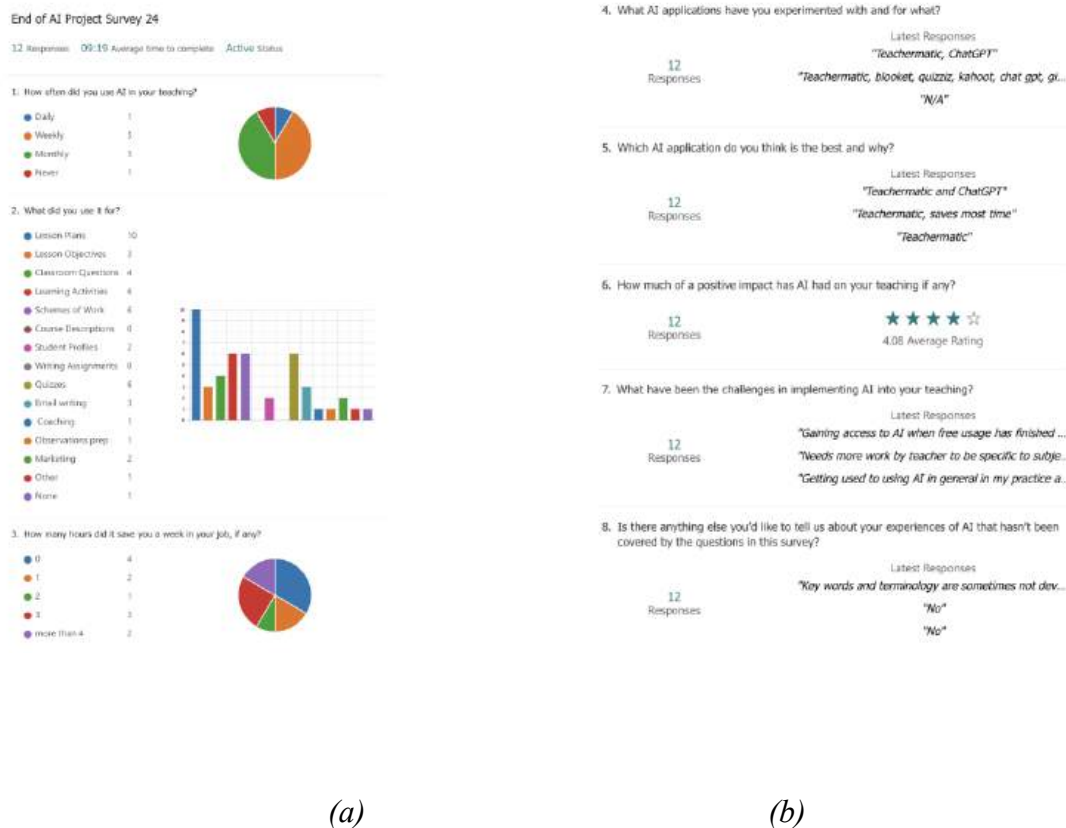


Figure. 32 (a, b): Visuals of End of AI Project Survey

TeacherMatic emerged as the most used and preferred tool, primarily for creating lesson plans, schemes of work, learning activities, and quizzes. Although some educators reported limited time savings, most savings were constricted between 0 to 2 hours per week. The overall impact

of AI on teaching is rated positively, with an average score of 4.08 out of 5. Challenges highlighted include limited access after free trials, the need for more subject-specific AI support, and general adaptation to AI tools. Despite these barriers, educators acknowledge the value of AI in improving efficiency and resource generation. Feedback also indicated a desire for clearer AI terminology and better integration support, pointing to the need for continued training and sustained access to effective AI tools.

TeacherMatic Review Survey

The TeacherMatic Review Survey provides valuable insights into the tool's usage, impact, and educators' opinions on its continued implementation. The responses reveal a mixed but generally positive perception, with some educators regularly using and benefiting from TeacherMatic, while others remain unsure or do not find it useful.

1. Frequency of Use

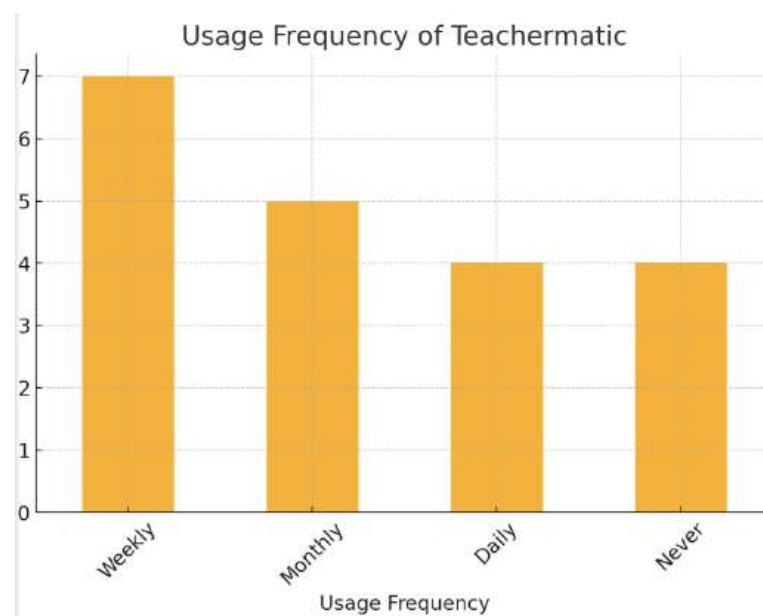


Figure 33: Usage Frequency of TeacherMatic

The responses indicate different levels of usage:

- Weekly users (7 respondents) from the largest group, showing that many educators include TeacherMatic into their teaching regularly.

- Monthly users (5 respondents) also demonstrated engagement, though less frequently.
- Daily users (4 respondents) represent a smaller but committed segment who rely heavily on the tool.
- Never users (4 respondents) highlight that some educators do not use TeacherMatic at all, possibly due to a lack of necessity, unfamiliarity, or inefficiencies.

This distribution suggests that TeacherMatic is a helpful tool for some but may not be essential or accessible for all educators.

2. Decision on Renewing the TeacherMatic License

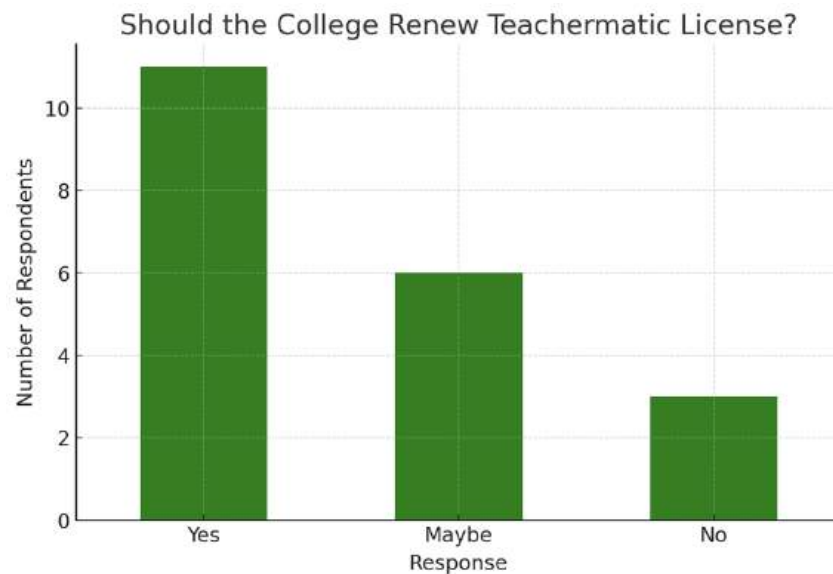


Figure 34: Survey decision on Renewing TeacherMatic License

When asked whether the college should renew its TeacherMatic license, responses were divided:

- Yes (11 respondents) – A majority supports renewal, indicating that many educators find it beneficial.
- Maybe (6 respondents) – A significant number remain uncertain, implying that they may not yet see a strong enough case for renewal.
- No (3 respondents) – A smaller group opposes renewal, suggesting dissatisfaction or preferences for alternatives.

The fact that a combined total of 9 respondents (Maybe + No) is uncertain or against renewal means that TeacherMatic may need improvements in functionality, trainings, or integration into teaching workflows.

3. Willingness to Continue Subscription

In addition to the institution's decision, educators were asked if they wanted to continue using TeacherMatic. The responses were:

- Yes (14 respondents) – The majority wish to maintain their access, indicating that they find value in the tool.
- No (6 respondents) – A notable minority does not wish to continue, suggesting that TeacherMatic may not meet their specific needs or that alternative tools are preferred.

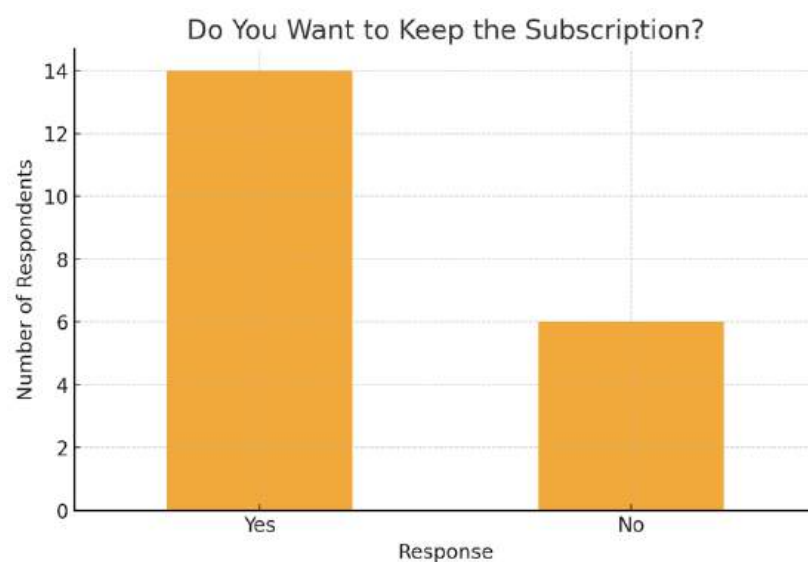


Figure 35: Survey data showing willingness to continue TeacherMatic Subscription

This indicated that while a strong majority finds the tool useful, there is a room to address concerns among those who are disengaged.

4. Perceived Positive Impact on Teaching

Respondents rated the positive impact of TeacherMatic on their teaching on a scale from 1 to 5, with an average rating of 3.15. This suggests:

- Moderate levels of satisfaction with TeacherMatic's impact.
- Some educators experience a clear benefit, but others may not see a strong enough effect on their teaching.

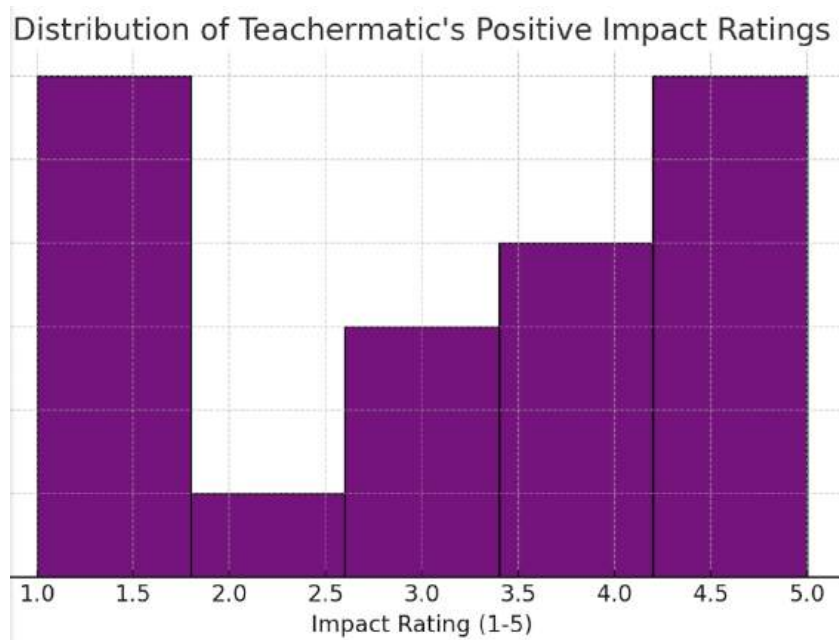


Figure 36: Impact of TeacherMatic on Teaching

This mixed rating suggests that while TeacherMatic is valuable for many, it may require refinements to enhance its effectiveness.

TeacherMatic Usage Data

07.01.25

- The increasing login trend suggests higher adoption of TeacherMatic.
- A large portion of users lack licences, possibly limiting access.
- Peaks in login activity suggest that usage correlates with training sessions or new feature rollouts.

This analysis helps institutions optimise AI tool adoption by tracking engagement, improving access, and tailoring training strategies for educators.

23.01.25

The data indicates growing adoption of TeacherMatic but inconsistent long-term engagement, emphasising the need for sustained user support and motivation.

13.02.25

The data reveals a mix of growing adoption and inconsistent engagement patterns, which may act as a barrier to full platform access and adoption. User engagement varies widely, where some users log in regularly while others experience long gaps, indicating inconsistent usage patterns. Sporadic login activity in the early months was followed by a sharp rise in late 2024 to early 2025. This surge suggests a possible link to institutional support, policy changes, or new feature rollouts, but engagement remains inconsistent over time.

Comparative Analysis of TeacherMatic Data

Comparative analysis evaluates TeacherMatic user activity across three datasets collected on 07 January 2025, 23 January 2025, and 13 February 2025. The primary focus is on user growth, license allocation, and login activity trends to understand how adoption and engagement evolve.

Growth in Total Users

One of the most notable findings is the steady increase in the total number of users from January to February 2025. A bar chart visualizing user count across these timestamps indicates a positive trend in platform adoption, suggesting more institutions or individuals are joining TeacherMatic.

Implications: A growing user base highlights successful outreach efforts or rising platform relevance in educational settings.

Potential Risks: If a decline in user count is observed in future datasets, it could indicate user disengagement, account deactivations, or a shift to competing platforms.

License Allocation Trend

The second key insight focuses on the distribution of allocated licenses over the three periods. A stacked bar chart reveals fluctuations in the proportion of licensed and unlicensed users. An increase in licensed users suggests more individuals gaining access to Teacher Matic premium features, reflecting successful adoption of paid or institutional licenses.

Implications: If many users remain unlicensed, they might have limited platform access, which could affect engagement. Monitoring how quickly new users transition from unlicensed to licensed is crucial for assessing the effectiveness of the licensing strategy.

Login Activity Trends and Engagement Patterns

Analysing login timespans through a line chart provides insights into user engagement patterns over time.

07 January 2025 Dataset: Sporadic logins with peaks on certain days, possibly linked to training sessions or institutional schedules.

23 January 2025 Dataset: A gradual increase in login frequency, indicating rising engagement levels.

13 February 2025 Datasets: A more consistent engagement pattern, with a higher number of users logging in regularly rather than sporadically.

Implications: Increasing logins suggest improving user engagement, potentially driven by feature rollouts, training, or institutional policies. A decline in logins could indicate user disinterest, preference for other tools, or competition from alternative platforms like Microsoft Copilot.

Key Challenges and Risks identified

User Retention: While user numbers are increasing, ensuring long-term engagement is critical.

License Distribution Efficiency: If unlicensed users remain high, accessibility barriers need to be addressed.

Competitor influence: The emergence of rival platforms like Microsoft Copilot could impact TeacherMatic's market position.

New Survey to Evaluate the Impact of AI-driven Large Language Models (LLMs) on Teaching Practices in FE

The survey, conducted across teaching staff in Further Education at NCC, intended to explore the adoption, perceived effectiveness, and challenges linked with AI tools in teaching practices. With 73 participants, the results provide both quantitative and qualitative insights into current usage trends and areas for development. An infographic summary of the survey is as below:

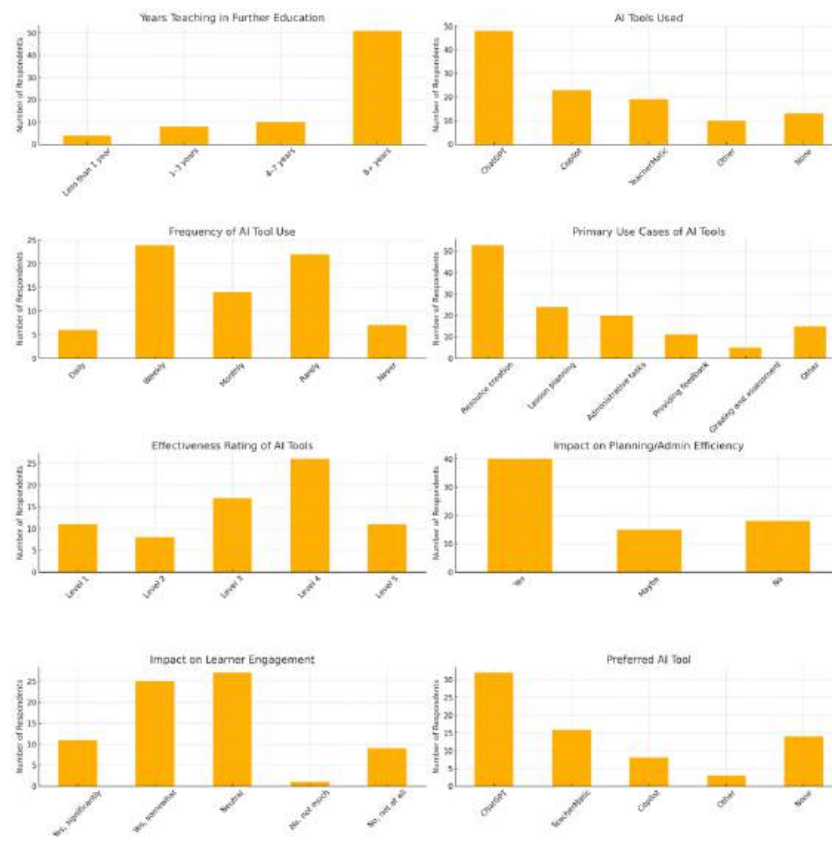


Figure 37: Infographic Summary of New Survey Data

Key Observations from the charts include:

Years of Teaching Experience: Most respondents (51 out of 73) have over 8 years of experience, meaning insights are from well-established professionals.

LLM used:

- ChatGPT is the most widely used AI tool (48 users).
- Copilot and TeacherMatic follow with moderate usage.
- 13 respondents do not use AI at all.

Frequency of AI Use: Most use AI on a weekly or rare basis, showing moderate integration into practice.

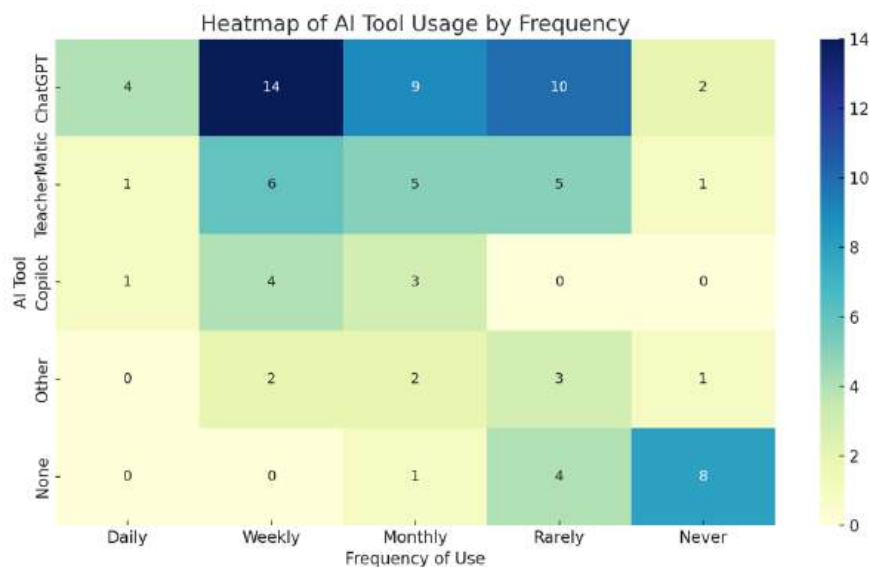


Figure 38: Chi-Square Test Heatmap of AI Usage by Frequency

LLM Use Cases:

- Resource creation leads by a large margin (53 respondents).
- Administrative tasks and lesson planning.

Perceived Effectiveness:

- Level 4 (out of 5) is the most common rating.
- The average sentiment is moderately positive.

Impact on Efficiency:

- 40 respondents (majority) report improved planning/admin efficiency.
- 15 are unsure and 18 saw no benefit.

Learner Engagement:

- About a half report positive impact.
- A large portion (27) remain neutral, and a small number reported no positive change.

Preferred AI tool:

- ChatGPT is again, the favourite.
- Some still prefer TeacherMatic, while a few remain undecided or prefer no tools.

The survey not only recognised key barriers to LLM adoption and doubts about content accuracy, but also found that lack of training, and data privacy were commonly cited challenges. 49% of the respondents voiced a clear interest in additional training, while another 37% were open to training depending on the format. This highlights the acute need for ongoing professional development and structured support to enhance confidence and competence in LLM usage.

In conclusion, the survey reveals an interestingly hopeful view of Large Language Models in education. ChatGPT, Copilot and TeacherMatic are not only being embraced for their practicality and time-saving capabilities, but their effective incorporation also depends on training, ethical use, and grouping with curriculum needs.

Chapter 5: Discussion and Recommendations

This chapter provides a critical interpretation of the findings presented in Chapter 4, situating them within the wider perspective of existing literature on Artificial Intelligence (AI), Large Language Model (LLMs), and their application in education, particularly Further Education (FE). It also explores the practical implications of the results, debates the limitations of the research, and provides evidence-based recommendations for educators, policymakers, and AI developers. The chapter concludes with proposals for future research directions, supporting the continuing dialogue about AI's responsible integration into educational practice.

Interpretation of Key Findings:

The fundamental goal of this research is to evaluate the impact of AI-driven Large Language Models (LLMs), such as TeacherMatic, ChatGPT and Microsoft Copilot on teaching practices in the FE sector. The study sought to define whether these tools enhance instructional delivery, planning efficiency, and learner engagement, while also identifying challenges educators face in AI adoption.

Adoption Trends and Usage Patterns: One of the clearest findings to appear is that AI tools—particularly general-purpose ones like ChatGPT have seen restrained but steadily increasing adoption among FE educators. ChatGPT is an extensively used tool, appreciated for its accessibility and versatility. In contrast, TeacherMatic, while receiving support, showed inconsistent usage patterns across users and NCC campuses.

The increase in TeacherMatic login frequency over the January-February 2025 period suggests increasing engagement, possibly influenced by institutional interventions such as training, policy encouragement, or trial periods. However, disparities in license allocations and the

notable portion of users remaining unlicensed highlighted systematic blocks to full adoption, suggesting that access equity, user onboarding, and license management strategies require attention.

Educator Perception and Effectiveness: Survey data revealed that most respondents perceived AI tools as moderately effective, with a mean effectiveness rating between 3.5 and 4.1 out of 5. These results show that AI-assisted resource generation significantly reduced cognitive and administrative load for educators. Similarly, teachers in the current study identified timesaving, resource support, and planning assistance as core benefits of tools like TeacherMatic. However, the efficiency is not universally acknowledged. For example, the TeacherMatic Review Survey reveals an average rating of 3.15, suggesting mixed satisfaction and strengthening the idea that the tool, while useful, may require improvements in adaptability, subject-specific functionality, and UX/UI design. Although many educators welcome AI for back-office efficiency, fewer opt for clear, measurable impacts on learner outcomes, and a substantial number remain unclear. The uncertainty highlights the need for longitudinal research into student engagement, attainment, and feedback in AI-enhanced learning environments.

Barriers to AI Integration: Several obstacles to AI integration are identified, echoing earlier studies by Huang (2023) and Holmes et. al. (2019). Educators expressed concerns about:

- Content accuracy and quality of AI outputs
- Lack of training or CPD opportunities
- Data Privacy and GDPR compliance
- Technical reliability and inconsistent performance

Evaluating LLM Integration Through the SAMR Model: To assess the pedagogical impact of LLMs like TeacherMatic, ChatGPT, and Copilot, the study applies the SAMR model, which categorises technology integration into four progressive stages as below:

- 1. Substitution - LLMs act as a direct tool substitute, with no functional change:** At the base level, many educators report using LLMs, particularly ChatGPT and TeacherMatic, for direct substitute of traditional tasks, such as generating lesson objectives, schemes of work, or drafting worksheets. For example, instead of manually writing a lesson plan, a teacher might input a topic into TeacherMatic and receive a ready-made structure. While this saves time, the fundamental pedagogical method remains unchanged, suggesting that many uses of LLMs are still operating within the substitution phase.
- 2. Augmentation - LLMs as functional improvements:** Various respondents describe using LLMs to improve existing processes. For instance, they employ LLMs to generate differentiated resources or tailor materials for mixed-ability groups, something that would have taken significantly longer manually. In these cases, LLMs enhance functionality by providing greater speed, adaptability, and scale, moving the tool's use into the augmentation

phase. This aligns with previous findings, which emphasise that AI-driven LLMs can meaningfully augment efficiency in educational settings without altering the underlying instructional strategies.

3. Modification - LLMs enable Significant Task Redesign: LLMs are helping with meaningful task redesign within FE settings, while the evidence of this move is still limited. Some teachers report using AI tools to facilitate class discussion, design role-play, and generate quizzes for formative assessments. These uses alter not just the speed but the nature of instructional tasks, embedding LLMs into more dynamic aspects of teaching and learning. However, these cases are less common, indicating that most educators have not yet crossed into this transformative level of integration.

4. Redefinition - LLMs allow for the creation of new tasks previously inconceivable: Only a few examples suggest redefinition-level practices. For instance, one respondent mentions using AI to co-design reflective learning journals with students, guiding learners to prompt ChatGPT for feedback on their work, under supervision. Another example includes using Copilot to automate content mapping against Ofsted criteria, streamlining compliance in a way that would have been unfeasible without LLMs. These innovative practices demonstrate the potential of LLMs to reshape teaching roles and learner agency but are currently exceptions rather than the norm.

SAMR Level	Example from	Commonality
Substitution	Using TeacherMatic	Very common
Augmentation	LLM-generated	Common
Modification	ChatGPT used for	Occasional
Redefinition	LLM supported	Rare

Table 5: Summary of LLMs Integration on the SAMR Model

Implications of SAMR Analysis

This analysis highlights that most current uses of LLMs in FE remain in the enhancement phase. While this still brings considerable benefits in terms of time efficiency, accessibility, and workload reduction, the transformative potential of LLMs, where teaching and teaching tasks are drastically reimaged, is yet to be broadly realised. The limited presence of Modification and Redefinition level use in this study suggests that educators may require more structured training, peer collaboration, and pedagogical experimentation to reach these higher levels in integration. Furthermore, redefinition level uses often depend on student digital literacy, institutional openness to innovation, and clear ethical frameworks, particularly when AI is used in concrete or evaluative roles. Without such support, LLMs are likely to remain a planning assistant rather than an educational change agent.

Relation to Existing Literature:

The findings of this study confirm, extend, and occasionally challenge earlier research on AI in education:

- The results support Estrellado et al. (2020) in affirming the practical benefits of data-driven tools in education. Tools like TeacherMatic, when used effectively, contribute to more streamlined planning and resource generation.
- The study builds on Gan et al. (2023) by providing empirical evidence from the UK FE sector that LLMs can enhance educator productivity, though not uniformly.
- However, unlike in higher education contexts where LLMs are often integrated into assessment and feedback according to Chen et al. (2023). FE practitioners in this study tend to limit LLMs' use to non-assessment-related tasks, suggesting institutional caution and perhaps a lack of confidence in AI reliability for high-stakes activities.
- Vygotsky's Zone of Proximal Development and the SAMR model have been used to evaluate the depth of LLM integration. Findings indicate that most current applications remain at the Substitution and Augmentation levels, which is helpful but not transformative. Only a few educators have reported the use of LLM that modified or redefined their teaching methods.

Limitations of the study:

The study is limited to a single FE institution, NCC, which may affect generalisability. While the college's multi-campus structure reflects some diversity, broader application of findings could be approached with caution. Those more interested in AI may have been more inclined to respond. The overall 12.15% survey response rate limits statistical power and may introduce non-response bias. Data relies on self-reported perceptions, which may not reflect actual usage or impact. The study focuses primarily on educators, excluding direct student perspectives. Additionally, the short timeframe due to the delay in project approval from the Education and Training Foundation (ETF) may not fully capture the long-term effects of LLM integration.

Practical Implications

The findings of this study carry significant implications for practice and policy within FE:

For Educators:

- LLMs can meaningfully reduce administrative workload and support planning efficiency.
- Effective use demands ongoing digital skills development and a reflective approach to tool integration
- Involvement in the co-design and feedback cycles of LLMs is important to ensure relevancy and usability

For Institutions:

- Must prioritise equitable access to LLMs, including timely license distribution and technical support.
- Investment in AI focused CPD, ideally differentiated by role and subject, is essential to support confident and ethical usage.
- LLM integration strategies should be aligned with curriculum goals and student-centred pedagogy.

For Developers and Policymakers:

- Tool developers should focus on usability, subject-specific customisation, and transparency of algorithms.
- Policymakers need to support ethical LLM adoption, ensuring compliance with data protection laws while encouraging innovation.
- There is a need for a National Framework for LLM in FE, incorporating ethical guidelines, technical standards, and professional development pathways.

Recommendations for Future Research:

Concluding the findings of this dissertation, future research on the implementation of Large Language Models (LLMs) in Further Education should use a broader lens to fully understand their impact. Multi-institutional studies are essential to examine how other educational settings use LLMs e.g. sixth form colleges, general FE colleges and adult education centres as funding, culture, resources and leadership can influence adoption. Longitudinal research should also be carried out to determine if LLMs affect key outcomes over time such as student retention, progression and achievement. In addition, student-focused research is required to explore their views on LLMs, engagement with AI and ethical concerns such as data privacy, model bias and dependence on automated content. Curriculum areas such as STEM may see the value and challenges of LLMs differently than creative arts and vocational subjects, making analyses difficult.

Furthermore, AI ethics in teacher training and digital strategies should be studied to ensure educators are competent and confident embracing this new AI landscape. As a ‘deeper think’ about the potential of AI and its ramifications on the education sector continue, close collaboration between the education and technology sectors is needed to harness both the potential and disruption of AI (Oriel Square, 2023). This will position AI as a supportive force in teaching and learning, enabling educators to enhance efficiency and foster inclusive learning environments.

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Appendices:

Survey Link: Evaluating the Impact of Large Language Model on teaching Practices in FE

<https://forms.office.com/e/YsuRmAn0n5>

Survey: Evaluating the Impact of Large Language Model on teaching Practices in FE

Evaluating the Impact of AI-driven Large Language Models on Teaching Practices in FE

15/03/2025, 11:25



Evaluating the Impact of AI-driven Large Language Models on Teaching Practices in FE

*"This survey aims to explore the role of AI tools such as **TeacherMatic**, **ChatGPT**, and **Copilot** in teaching practices within UK Further Education (FE). Your responses will help assess the effectiveness, challenges, and future potential of AI-powered tools in lesson planning, resource creation, and student engagement. The survey should take approximately **10-15 minutes** to complete. Your responses will remain anonymous, **confidential** and will be used solely for research purposes."*

Thank You for your valuable contribution!

Demographics & Teaching Experience



<https://forms.office.com/Pages/DesignPage.aspx?from=ghr...3e5d0058LFI6Mj...L...4UQ1Q3TExu8FYV54e5722Dv7N8L4M56Y4e> Page 1 of 14

1.

How many years have you been teaching in Further Education? *

- ☐ Less than 1 year
- ☐ 1-3 years
- ☐ 4-7 years
- ☐ 8+ years

2.

What is your curriculum area? *

- ☐ Maths
- ☐ English
- ☐ Business
- ☐ ESOL
- ☐ ICT & Creative
- ☐ Sports
- ☐ SEND
- ☐ Apprenticeships

- ☐ Health & Early Years
- ☐ Science
- ☐ Higher Education
- ☐ Teacher Education
- ☐ Hair & Beauty
- ☐ Construction & Engineering
- ☐ Hospitality & Catering
- ☐ A Level Academy
- ☐ Other

AI Tool Usage in Teaching



3.

Which AI tool(s) have you used for lesson planning, resource creation, or teaching support? *

- ☐ TeacherMatic
- ☐ ChatGPT
- ☐ Copilot
- ☐ Other
- ☐ I have not used any AI tools in my teaching

4.

What specific tool have you used if you answered 'other' to Q3? *

5.

How often do you use AI tools for teaching-related tasks? *

- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Rarely
- ☐ Never

6

For which tasks do you primarily use AI tools? *

- ☐ Lesson planning
- ☐ Resource creation (e.g., worksheets, quizzes)
- ☐ Grading and assessment
- ☐ Providing feedback to students
- ☐ Administrative tasks
- ☐ Other

Perceived Effectiveness of AI Tools



7

How effective do you find AI tools in enhancing your teaching? (Scale = 1*Less effective - 5* Most effective) *

1	2	3	4	5
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8

Please explain your rating. *

9

In what ways have AI tools impacted your teaching experience? *

10

Have the AI tools impacted your planning, admin tasks, and efficiency? *

- ☐ Yes
- ☐ No
- ☐ Maybe

11

Do you believe AI tools (like TeacherMatic) have positively impacted your learner engagement and their learning outcomes? *

- ☐ Yes, significantly
- ☐ Yes, somewhat
- ☐ Neutral
- ☐ No, not much
- ☐ No, not at all

12

Can you provide examples to explain your answer to Q9 and how the use of AI affected your students (positively or negatively)? *

Challenges & Barriers to AI Adoption



14

What improvements to the technology would make AI tools more useful in your teaching? ⁴

13

What challenges have you faced when using AI tools for teaching? Please rank. *

Lack of training or familiarity with AI tools
Concerns about accuracy and reliability of AI-generated content
Data privacy and security concerns
Resistance from colleagues or institutions
AI-generated content lacks human creativity/personal touch
Technical difficulties or access issues

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Preferences & Future Use of AI in Education



15

Which AI tool do you prefer for teaching-related tasks? *

- ☐ TeacherMatic
- ☐ ChatGPT
- ☐ Copilot
- ☐ Other
- ☐ I do not use AI tools

16

Would you recommend that your institution **renew its TeacherMatic subscription?** *

- ☐ Yes
- ☐ No

17

Would you like more training on how to effectively use AI tools in Further Education? *

- ☐ Yes, I would find training beneficial
- ☐ No, I am comfortable using AI tools
- ☐ Maybe, depending on the training format and content

18

What are your thoughts on the **future role of AI in education?** *

Final Comments



19

Do you have any additional feedback or suggestions regarding AI tools in Further Education? *

This content is neither created nor endorsed by Microsoft. The data you submit will be sent to the form owner.



New Survey Data: Evaluating the Impact of AI-driven Large Language Model LLMs on Teaching Practices in FE

Survey data collected of respondents from Redbridge, Havering, Rainham and Poplar campuses at New City College.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50	Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60	Q61	Q62	Q63	Q64	Q65	Q66	Q67	Q68	Q69	Q70	Q71	Q72	Q73	Q74	Q75	Q76	Q77	Q78	Q79	Q80	Q81	Q82	Q83	Q84	Q85	Q86	Q87	Q88	Q89	Q90	Q91	Q92	Q93	Q94	Q95	Q96	Q97	Q98	Q99	Q100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100

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Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40	Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50	Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60	Q61	Q62	Q63	Q64	Q65	Q66	Q67	Q68	Q69	Q70	Q71	Q72	Q73	Q74	Q75	Q76	Q77	Q78	Q79	Q80	Q81	Q82	Q83	Q84	Q85	Q86	Q87	Q88	Q89	Q90	Q91	Q92	Q93	Q94	Q95	Q96	Q97	Q98	Q99	Q100
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1. What type
 2. What are the interesting uses of generating information in order to
 3. Can computers make any predictions for the future and, if so, how good are these predictions?
 4. What can be done to make them better and control the harmful results? (It is the same good suggestions for future)
 5. definition of the following
 6. Required: identify
 7. the phases of the
 8. in a defined area
 9. how
 10. the two stages and explain in detail how they differ from each other
 11. What is a point in time analysis?
 12. What are the benefits of this analysis? What are the main reasons for using it?
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(h)

1. **What are the main differences between the two types of research?**
 The first type of research is **qualitative research**, which is concerned with understanding the meaning and experience of phenomena. The second type of research is **quantitative research**, which is concerned with measuring and testing hypotheses.

2. **What are the main strengths and weaknesses of each type of research?**
 Qualitative research has the strength of providing rich, detailed data, but it is often subjective and difficult to generalize. Quantitative research has the strength of providing objective, measurable data, but it can be less detailed and less flexible.

3. **What are the main ethical considerations in research?**
 The main ethical considerations in research are **informed consent**, **confidentiality**, and **minimizing harm**.

4. **What are the main steps in the research process?**
 The main steps in the research process are **identifying a research problem**, **developing a research plan**, **collecting data**, **analyzing data**, and **reporting results**.

5. **What are the main challenges in research?**
 The main challenges in research are **time and resource constraints**, **complexity of phenomena**, and **subjectivity of interpretation**.

6. **What are the main applications of research?**
 The main applications of research are **improving practice**, **developing theory**, and **informing policy**.

7. **What are the main future directions of research?**
 The main future directions of research are **increasing rigor**, **improving transparency**, and **enhancing collaboration**.

8. **What are the main conclusions of the research?**
 The main conclusions of the research are that **research is a complex and challenging process**, but it is also a **valuable and rewarding activity**.

9. **What are the main implications of the research?**
 The main implications of the research are that **research should be conducted with care and attention**, and that **research should be used to inform practice and policy**.

10. **What are the main recommendations of the research?**
 The main recommendations of the research are that **research should be conducted with rigor and integrity**, and that **research should be used to improve practice and policy**.

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 52. *Myself* (100%)
 53. *Myself* (100%)
 54. *Myself* (100%)
 55. *Myself* (100%)
 56. *Myself* (100%)
 57. *Myself* (100%)
 58. *Myself* (100%)
 59. *Myself* (100%)
 60. *Myself* (100%)
 61. *Myself* (100%)
 62. *Myself* (100%)
 63. *Myself* (100%)
 64. *Myself* (100%)
 65. *Myself* (100%)
 66. *Myself* (100%)
 67. *Myself* (100%)
 68. *Myself* (100%)
 69. *Myself* (100%)
 70. *Myself* (100%)
 71. *Myself* (100%)
 72. *Myself* (100%)
 73. *Myself* (100%)
 74. *Myself* (100%)
 75. *Myself* (100%)
 76. *Myself* (100%)
 77. *Myself* (100%)
 78. *Myself* (100%)
 79. *Myself* (100%)
 80. *Myself* (100%)
 81. *Myself* (100%)
 82. *Myself* (100%)
 83. *Myself* (100%)
 84. *Myself* (100%)
 85. *Myself* (100%)
 86. *Myself* (100%)
 87. *Myself* (100%)
 88. *Myself* (100%)
 89. *Myself* (100%)
 90. *Myself* (100%)
 91. *Myself* (100%)
 92. *Myself* (100%)
 93. *Myself* (100%)
 94. *Myself* (100%)
 95. *Myself* (100%)
 96. *Myself* (100%)
 97. *Myself* (100%)
 98. *Myself* (100%)
 99. *Myself* (100%)
 100. *Myself* (100%)

(j)

[illegible][illegible]

(1)

[illegible]

(n)

[illegible]

(p)

R2105D12222502

<https://forms.office.com/Pages/AnalysisPage.aspx?AnalyzerToken=JehR6zztxnmOyhDYwoOXMfSTWTto5UKW&id=mUft038080m159qCPJXo2dCgwrvkEeFOmcU6rNwOUPhUMEk0QTVTQVJQU1k5VE1TVUVQUjU0VjNZNy4u>

The following table represents the content of the spreadsheet shown in the screenshots:

Question	Answer
1. How many Technomath's Review Survey questions did you answer?	20
2. How many Technomath's Review Survey questions did you answer?	20
3. How many Technomath's Review Survey questions did you answer?	20
4. How many Technomath's Review Survey questions did you answer?	20
5. How many Technomath's Review Survey questions did you answer?	20
6. How many Technomath's Review Survey questions did you answer?	20
7. How many Technomath's Review Survey questions did you answer?	20
8. How many Technomath's Review Survey questions did you answer?	20
9. How many Technomath's Review Survey questions did you answer?	20
10. How many Technomath's Review Survey questions did you answer?	20
11. How many Technomath's Review Survey questions did you answer?	20
12. How many Technomath's Review Survey questions did you answer?	20
13. How many Technomath's Review Survey questions did you answer?	20
14. How many Technomath's Review Survey questions did you answer?	20
15. How many Technomath's Review Survey questions did you answer?	20
16. How many Technomath's Review Survey questions did you answer?	20
17. How many Technomath's Review Survey questions did you answer?	20
18. How many Technomath's Review Survey questions did you answer?	20
19. How many Technomath's Review Survey questions did you answer?	20
20. How many Technomath's Review Survey questions did you answer?	20
21. How many Technomath's Review Survey questions did you answer?	20
22. How many Technomath's Review Survey questions did you answer?	20
23. How many Technomath's Review Survey questions did you answer?	20
24. How many Technomath's Review Survey questions did you answer?	20
25. How many Technomath's Review Survey questions did you answer?	20
26. How many Technomath's Review Survey questions did you answer?	20
27. How many Technomath's Review Survey questions did you answer?	20
28. How many Technomath's Review Survey questions did you answer?	20
29. How many Technomath's Review Survey questions did you answer?	20
30. How many Technomath's Review Survey questions did you answer?	20

TeacherMatic Usage Data 07.01.2025

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'R2105D1222502', with columns for 'Name', 'Date', 'Time', 'Status', 'Type', 'Device', and 'Usage'. The data is organized in a grid format with multiple rows and columns.

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'R2105D1222502', with columns for 'Name', 'Date', 'Time', 'Status', 'Type', 'Device', and 'Usage'. The data is organized in a grid format with multiple rows and columns.

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'R2105D1222502', with columns for 'Name', 'Date', 'Time', 'Status', 'Type', 'Device', and 'Usage'. The data is organized in a grid format with multiple rows and columns.

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'R2105D1222502', with columns for 'Name', 'Date', 'Time', 'Status', 'Type', 'Device', and 'Usage'. The data is organized in a grid format with multiple rows and columns.

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'R2105D1222502', with columns for 'Name', 'Date', 'Time', 'Status', 'Type', 'Device', and 'Usage'. The data is organized in a grid format with multiple rows and columns.

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'R2105D1222502', with columns for 'Name', 'Date', 'Time', 'Status', 'Type', 'Device', and 'Usage'. The data is organized in a grid format with multiple rows and columns.

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'R2105D1222502', with columns for 'Name', 'Date', 'Time', 'Status', 'Type', 'Device', and 'Usage'. The data is organized in a grid format with multiple rows and columns.

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'R2105D1222502', with columns for 'Name', 'Date', 'Time', 'Status', 'Type', 'Device', and 'Usage'. The data is organized in a grid format with multiple rows and columns.

R2105D1222502

m

TeacherMatic Usage Data 23.01.2025

The screenshot displays the Microsoft Excel 2010 application window. The title bar indicates the file name is 'Book1 - Microsoft Excel'. The ribbon is set to the 'Formulas' tab, with the 'Calculation Options' group expanded, showing 'Automatic' as the selected option. The 'Formulas' tab includes buttons for 'AutoSum', 'Insert Function', 'Calculate Now', 'Calculate Using Table', 'Evaluate Formula', and 'Show Formulas'. The 'Formulas' pane on the right shows the 'Name Box' with 'A1' selected and the 'Formula Bar' with '=A1'. The spreadsheet grid shows column A containing a list of names and dates, such as 'John Doe 1/1/2010', 'Jane Smith 2/1/2010', and 'John Doe 3/1/2010'. The rest of the spreadsheet is obscured by a large black redaction box.

The screenshot shows a Windows 10 desktop. In the foreground, a web browser window is open, displaying a list of IP addresses. The list is organized into columns, with the first column containing a large number of IP addresses, and subsequent columns containing smaller numbers. The IP addresses are listed in a single column, with some addresses appearing multiple times. The list includes addresses such as 192.168.1.1, 192.168.1.2, 192.168.1.3, 192.168.1.4, 192.168.1.5, 192.168.1.6, 192.168.1.7, 192.168.1.8, 192.168.1.9, 192.168.1.10, 192.168.1.11, 192.168.1.12, 192.168.1.13, 192.168.1.14, 192.168.1.15, 192.168.1.16, 192.168.1.17, 192.168.1.18, 192.168.1.19, 192.168.1.20, 192.168.1.21, 192.168.1.22, 192.168.1.23, 192.168.1.24, 192.168.1.25, 192.168.1.26, 192.168.1.27, 192.168.1.28, 192.168.1.29, 192.168.1.30, 192.168.1.31, 192.168.1.32, 192.168.1.33, 192.168.1.34, 192.168.1.35, 192.168.1.36, 192.168.1.37, 192.168.1.38, 192.168.1.39, 192.168.1.40, 192.168.1.41, 192.168.1.42, 192.168.1.43, 192.168.1.44, 192.168.1.45, 192.168.1.46, 192.168.1.47, 192.168.1.48, 192.168.1.49, 192.168.1.50, 192.168.1.51, 192.168.1.52, 192.168.1.53, 192.168.1.54, 192.168.1.55, 192.168.1.56, 192.168.1.57, 192.168.1.58, 192.168.1.59, 192.168.1.60, 192.168.1.61, 192.168.1.62, 192.168.1.63, 192.168.1.64, 192.168.1.65, 192.168.1.66, 192.168.1.67, 192.168.1.68, 192.168.1.69, 192.168.1.70, 192.168.1.71, 192.168.1.72, 192.168.1.73, 192.168.1.74, 192.168.1.75, 192.168.1.76, 192.168.1.77, 192.168.1.78, 192.168.1.79, 192.168.1.80, 192.168.1.81, 192.168.1.82, 192.168.1.83, 192.168.1.84, 192.168.1.85, 192.168.1.86, 192.168.1.87, 192.168.1.88, 192.168.1.89, 192.168.1.90, 192.168.1.91, 192.168.1.92, 192.168.1.93, 192.168.1.94, 192.168.1.95, 192.168.1.96, 192.168.1.97, 192.168.1.98, 192.168.1.99, 192.168.1.100, 192.168.1.101, 192.168.1.102, 192.168.1.103, 192.168.1.104, 192.168.1.105, 192.168.1.106, 192.168.1.107, 192.168.1.108, 192.168.1.109, 192.168.1.110, 192.168.1.111, 192.168.1.112, 192.168.1.113, 192.168.1.114, 192.168.1.115, 192.168.1.116, 192.168.1.117, 192.168.1.118, 192.168.1.119, 192.168.1.120, 192.168.1.121, 192.168.1.122, 192.168.1.123, 192.168.1.124, 192.168.1.125, 192.168.1.126, 192.168.1.127, 192.168.1.128, 192.168.1.129, 192.168.1.130, 192.168.1.131, 192.168.1.132, 192.168.1.133, 192.168.1.134, 192.168.1.135, 192.168.1.136, 192.168.1.137, 192.168.1.138, 192.168.1.139, 192.168.1.140, 192.168.1.141, 192.168.1.142, 192.168.1.143, 192.168.1.144, 192.168.1.145, 192.168.1.146, 192.168.1.147, 192.168.1.148, 192.168.1.149, 192.168.1.150, 192.168.1.151, 192.168.1.152, 192.168.1.153, 192.168.1.154, 192.168.1.155, 192.168.1.156, 192.168.1.157, 192.168.1.158, 192.168.1.159, 192.168.1.160, 192.168.1.161, 192.168.1.162, 192.168.1.163, 192.168.1.164, 192.168.1.165, 192.168.1.166, 192.168.1.167, 192.168.1.168, 192.168.1.169, 192.168.1.170, 192.168.1.171, 192.168.1.172, 192.168.1.173, 192.168.1.174, 192.168.1.175, 192.168.1.176, 192.168.1.177, 192.168.1.178, 192.168.1.179, 192.168.1.180, 192.168.1.181, 192.168.1.182, 192.168.1.183, 192.168.1.184, 192.168.1.185, 192.168.1.186, 192.168.1.187, 192.168.1.188, 192.168.1.189, 192.168.1.190, 192.168.1.191, 192.168.1.192, 192.168.1.193, 192.168.1.194, 192.168.1.195, 192.168.1.196, 192.168.1.197, 192.168.1.198, 192.168.1.199, 192.168.1.200, 192.168.1.201, 192.168.1.202, 192.168.1.203, 192.168.1.204, 192.168.1.205, 192.168.1.206, 192.168.1.207, 192.168.1.208, 192.168.1.209, 192.168.1.210, 192.168.1.211, 192.168.1.212, 192.168.1.213, 192.168.1.214, 192.168.1.215, 192.168.1.216, 192.168.1.217, 192.168.1.218, 192.168.1.219, 192.168.1.220, 192.168.1.221, 192.168.1.222, 192.168.1.223, 192.168.1.224, 192.168.1.225, 192.168.1.226, 192.168.1.227, 192.168.1.228, 192.168.1.229, 192.168.1.230, 192.168.1.231, 192.168.1.232, 192.168.1.233, 192.168.1.234, 192.168.1.235, 192.168.1.236, 192.168.1.237, 192.168.1.238, 192.168.1.239, 192.168.1.240, 192.168.1.241, 192.168.1.242, 192.168.1.243, 192.168.1.244, 192.168.1.245, 192.168.1.246, 192.168.1.247, 192.168.1.248, 192.168.1.249, 192.168.1.250, 192.168.1.251, 192.168.1.252, 192.168.1.253, 192.168.1.254, 192.168.1.255.

The screenshot shows the Microsoft Excel 2010 application window. The title bar indicates the file is 'Book1 - Microsoft Excel'. The ribbon is set to the 'Formulas' tab, with the 'Date & Time' group selected. The formula bar displays the formula '=DATE(2012,6,1)'. The worksheet contains a list of dates in column A, starting from 6/1/2012 and ending on 6/30/2012. The dates are listed in a column that is 10 columns wide, with the date values appearing in the first column of the range.

Date
6/1/2012
6/2/2012
6/3/2012
6/4/2012
6/5/2012
6/6/2012
6/7/2012
6/8/2012
6/9/2012
6/10/2012
6/11/2012
6/12/2012
6/13/2012
6/14/2012
6/15/2012
6/16/2012
6/17/2012
6/18/2012
6/19/2012
6/20/2012
6/21/2012
6/22/2012
6/23/2012
6/24/2012
6/25/2012
6/26/2012
6/27/2012
6/28/2012
6/29/2012
6/30/2012

The screenshot shows the Microsoft Excel application window. The title bar indicates the file is named 'Date' and is located on the 'Date' tab of a workbook. The ribbon is set to the 'Formulas' tab, with the 'AutoFill' dropdown menu open, showing options like 'Fill Down', 'Fill Right', 'Fill Left', and 'Fill Up'. The main grid shows a list of dates in column A, starting from 2012-01-01 and ending on 2012-01-31. The active cell is A1, which contains the date 2012-01-01. The status bar at the bottom indicates the active cell is A1 and the formula bar is empty.

The screenshot shows a Microsoft Excel spreadsheet with the following data:

IP-Adresse	Host
192.168.1.1	192.168.1.1
192.168.1.2	192.168.1.2
192.168.1.3	192.168.1.3
192.168.1.4	192.168.1.4
192.168.1.5	192.168.1.5
192.168.1.6	192.168.1.6
192.168.1.7	192.168.1.7
192.168.1.8	192.168.1.8
192.168.1.9	192.168.1.9
192.168.1.10	192.168.1.10
192.168.1.11	192.168.1.11
192.168.1.12	192.168.1.12
192.168.1.13	192.168.1.13
192.168.1.14	192.168.1.14
192.168.1.15	192.168.1.15
192.168.1.16	192.168.1.16
192.168.1.17	192.168.1.17
192.168.1.18	192.168.1.18
192.168.1.19	192.168.1.19
192.168.1.20	192.168.1.20
192.168.1.21	192.168.1.21
192.168.1.22	192.168.1.22
192.168.1.23	192.168.1.23
192.168.1.24	192.168.1.24
192.168.1.25	192.168.1.25
192.168.1.26	192.168.1.26
192.168.1.27	192.168.1.27
192.168.1.28	192.168.1.28
192.168.1.29	192.168.1.29
192.168.1.30	192.168.1.30
192.168.1.31	192.168.1.31
192.168.1.32	192.168.1.32
192.168.1.33	192.168.1.33
192.168.1.34	192.168.1.34
192.168.1.35	192.168.1.35
192.168.1.36	192.168.1.36
192.168.1.37	192.168.1.37
192.168.1.38	192.168.1.38
192.168.1.39	192.168.1.39
192.168.1.40	192.168.1.40
192.168.1.41	192.168.1.41
192.168.1.42	192.168.1.42
192.168.1.43	192.168.1.43
192.168.1.44	192.168.1.44
192.168.1.45	192.168.1.45
192.168.1.46	192.168.1.46
192.168.1.47	192.168.1.47
192.168.1.48	192.168.1.48
192.168.1.49	192.168.1.49
192.168.1.50	192.168.1.50
192.168.1.51	192.168.1.51
192.168.1.52	192.168.1.52
192.168.1.53	192.168.1.53
192.168.1.54	192.168.1.54
192.168.1.55	192.168.1.55
192.168.1.56	192.168.1.56
192.168.1.57	192.168.1.57
192.168.1.58	192.168.1.58
192.168.1.59	192.168.1.59
192.168.1.60	192.168.1.60
192.168.1.61	192.168.1.61
192.168.1.62	192.168.1.62
192.168.1.63	192.168.1.63
192.168.1.64	192.168.1.64
192.168.1.65	192.168.1.65
192.168.1.66	192.168.1.66
192.168.1.67	192.168.1.67
192.168.1.68	192.168.1.68
192.168.1.69	192.168.1.69
192.168.1.70	192.168.1.70
192.168.1.71	192.168.1.71
192.168.1.72	192.168.1.72
192.168.1.73	192.168.1.73
192.168.1.74	192.168.1.74
192.168.1.75	192.168.1.75
192.168.1.76	192.168.1.76
192.168.1.77	192.168.1.77
192.168.1.78	192.168.1.78
192.168.1.79	192.168.1.79
192.168.1.80	192.168.1.80
192.168.1.81	192.168.1.81
192.168.1.82	192.168.1.82
192.168.1.83	192.168.1.83
192.168.1.84	192.168.1.84
192.168.1.85	192.168.1.85
192.168.1.86	192.168.1.86
192.168.1.87	192.168.1.87
192.168.1.88	192.168.1.88
192.168.1.89	192.168.1.89
192.168.1.90	192.168.1.90
192.168.1.91	192.168.1.91
192.168.1.92	192.168.1.92
192.168.1.93	192.168.1.93
192.168.1.94	192.168.1.94
192.168.1.95	192.168.1.95
192.168.1.96	192.168.1.96
192.168.1.97	192.168.1.97
192.168.1.98	192.168.1.98
192.168.1.99	192.168.1.99
192.168.1.100	192.168.1.100
192.168.1.101	19

The screenshot shows the Microsoft Excel application window. The title bar at the top reads "Microsoft Excel - [workbook1.xlsx]". The ribbon is set to the "Formulas" tab, with the "Formula Auditing" group selected. The spreadsheet area contains a large black rectangular area on the left side, which appears to be a redaction or a placeholder for an image. The rest of the spreadsheet is empty, showing the standard grid with columns A through Z and rows 1 through 1048576. The status bar at the bottom indicates "Formulas" and "Active Cell: A1".

TeacherMatic Usage Data 13.02.2025

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'Admin', 'Teacher', and 'Student', along with their respective usage statistics. The interface includes a menu bar at the top and a toolbar with icons for different functions.

User	Usage	Access	Comments
Admin	100%	100%	
Teacher	100%	100%	
Student	100%	100%	

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'Admin', 'Teacher', and 'Student', along with their respective usage statistics. The interface includes a menu bar at the top and a toolbar with icons for different functions.

User	Usage	Access	Comments
Admin	100%	100%	
Teacher	100%	100%	
Student	100%	100%	

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'Admin', 'Teacher', and 'Student', along with their respective usage statistics. The interface includes a menu bar at the top and a toolbar with icons for different functions.

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Teacher	100%	100%	
Student	100%	100%	

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Admin	100%	100%	
Teacher	100%	100%	
Student	100%	100%	

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'Admin', 'Teacher', and 'Student', along with their respective usage statistics. The interface includes a menu bar at the top and a toolbar with icons for different functions.

User	Usage	Access	Comments
Admin	100%	100%	
Teacher	100%	100%	
Student	100%	100%	

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'Admin', 'Teacher', and 'Student', along with their respective usage statistics. The interface includes a menu bar at the top and a toolbar with icons for different functions.

User	Usage	Access	Comments
Admin	100%	100%	
Teacher	100%	100%	
Student	100%	100%	

This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'Admin', 'Teacher', and 'Student', along with their respective usage statistics. The interface includes a menu bar at the top and a toolbar with icons for different functions.

User	Usage	Access	Comments
Admin	100%	100%	
Teacher	100%	100%	
Student	100%	100%	

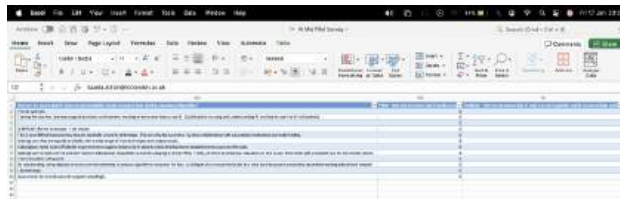
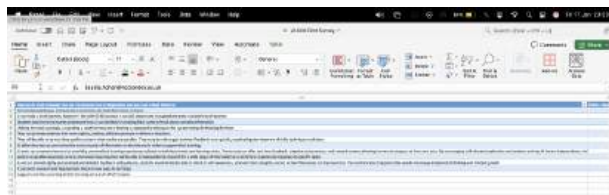
This screenshot shows the 'Users' tab in the TeacherMatic software. The table lists various users, including 'Admin', 'Teacher', and 'Student', along with their respective usage statistics. The interface includes a menu bar at the top and a toolbar with icons for different functions.

User	Usage	Access	Comments
Admin	100%	100%	
Teacher	100%	100%	
Student	100%	100%	

R2105D1222502

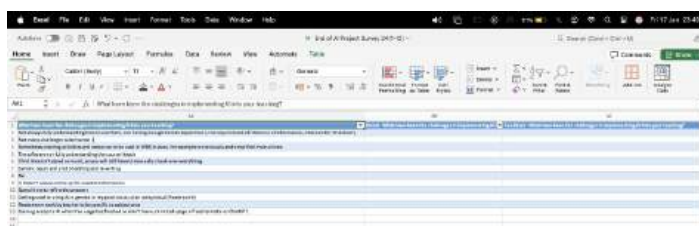
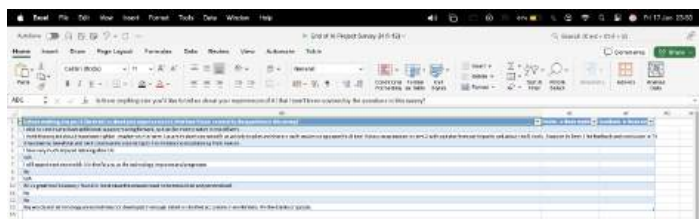
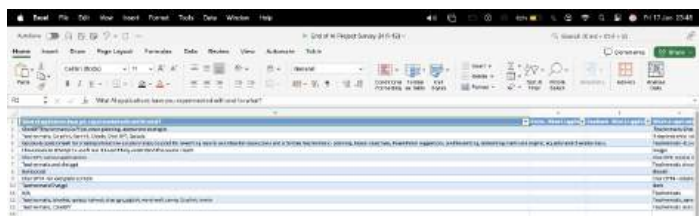
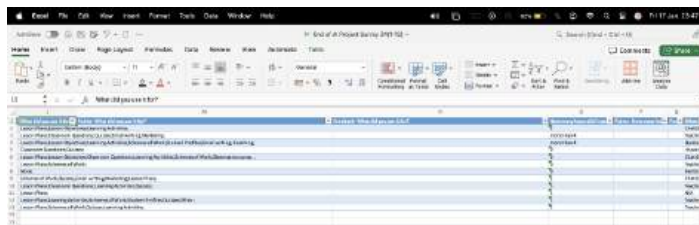
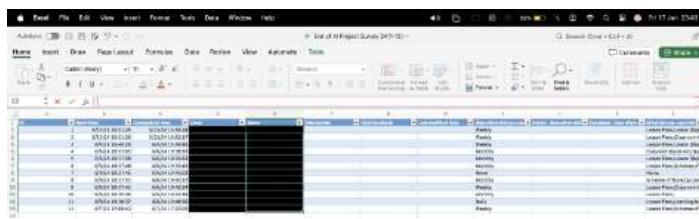


q



End of AI Project Survey Data (June 2024):

R2105D12222502



R2105D1222502